

Substrate Training Course Driscolls Morocco (Nov 2020)

Module 2a Plant Water Use +Irrigation



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Topics Covered Today

Section 1- Plant Water Use and Transpiration

- The role of water in plants
- The Importance of Transpiration
- How is Water Taken up and Transported by Plants
- What factors affect water uptake & transpiration?

Section 2 – Substrate Physical Properties

- Air/Water Relations in Substrate
- Understanding Matric Potential (Moisture Tension)
- The make-up of Peat and Coir Substrates
- Adjusting Air/Water Holding Capacity of substrates

Section 3 – Assessing Substrate Moisture Content and Run Off

- Methods of Assessing Substrate Moisture Content
- Run off – Why its necessary & how to measure it?
- Adjustment of Substrate Moisture + Run Off at different crop Growth Stages.

Section 4 – Crop Water Requirement & Irrigation Scheduling

- Meeting Hourly ET Demand
- Irrigation Scheduling in Substrate Grown Crops.
- Peak Water Use by Strawberry, Raspberry & Blueberries in Substrate.



Table Top Strawberries



Victoria L-Canes

Section 1

Plant Water Use and Transpiration

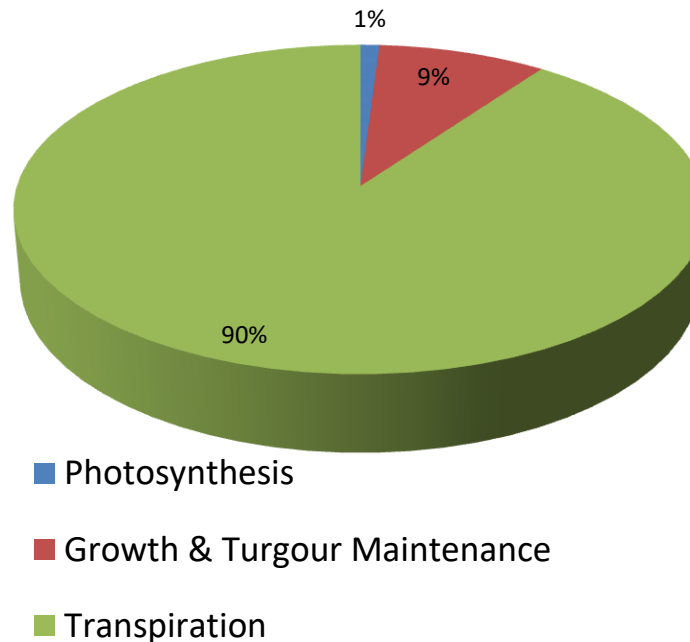


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Water Use by Berry Plants

Typical Water use by most plants:

- 1 % = Photosynthesis
- 9 % = Growth + Turgour Maintenance
(Cell division/cell expansion)
- **90 = % Transpiration**



Thus, for every 1 litre of water taken up by our crops:

- 900 ml is used in transpiration.



What is Transpiration?

Transpiration is the movement of water:

- from roots, through the Xylem (water transport) vessels
 - to different parts of the plant, for:
- 1. Supply of Water**
 - to developing cells/organs (for growth) & for Photosynthesis.
 - 2. Transport of some nutrients** (especially Calcium & Boron)
 - to developing cells/organs.
 - Most other nutrients can be transported in cell sap (Phloem).
 - 3. Cooling of the plant**
 - by evaporation of water through pores in the leaf (stomata).

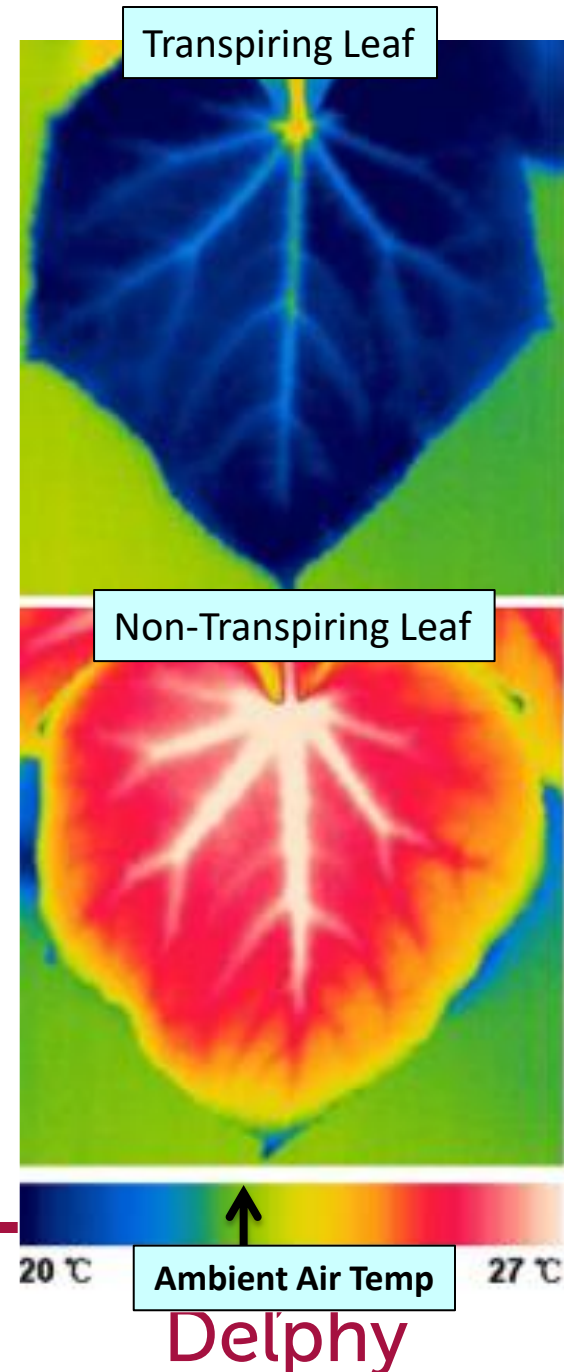
An actively transpiring plant leaf is **4-5°C cooler** than the surrounding air.

A non-transpiring (stressed) leaf heats up

- Stress can be from: dry pots, high VPD, high wind speed, crop damage.
- Stressed leaves can be **5-7°C warmer** than the surrounding air.
- (plant being cooked in its own juices!).

Thermal Imaging cameras can pictorially show plant stress:

Thermal Images of Transpiring & Non- Transpiring Cucumber Leaves



How do Plants Absorb Water in Soil/Substrate?

Substrates are a mix of loosely packed particles.

- Between the particles are air spaces (pores).
- Pores can be full of air **or** water – depending on
- their size and the substrate moisture content.

Pore water is absorbed' by **root hairs**

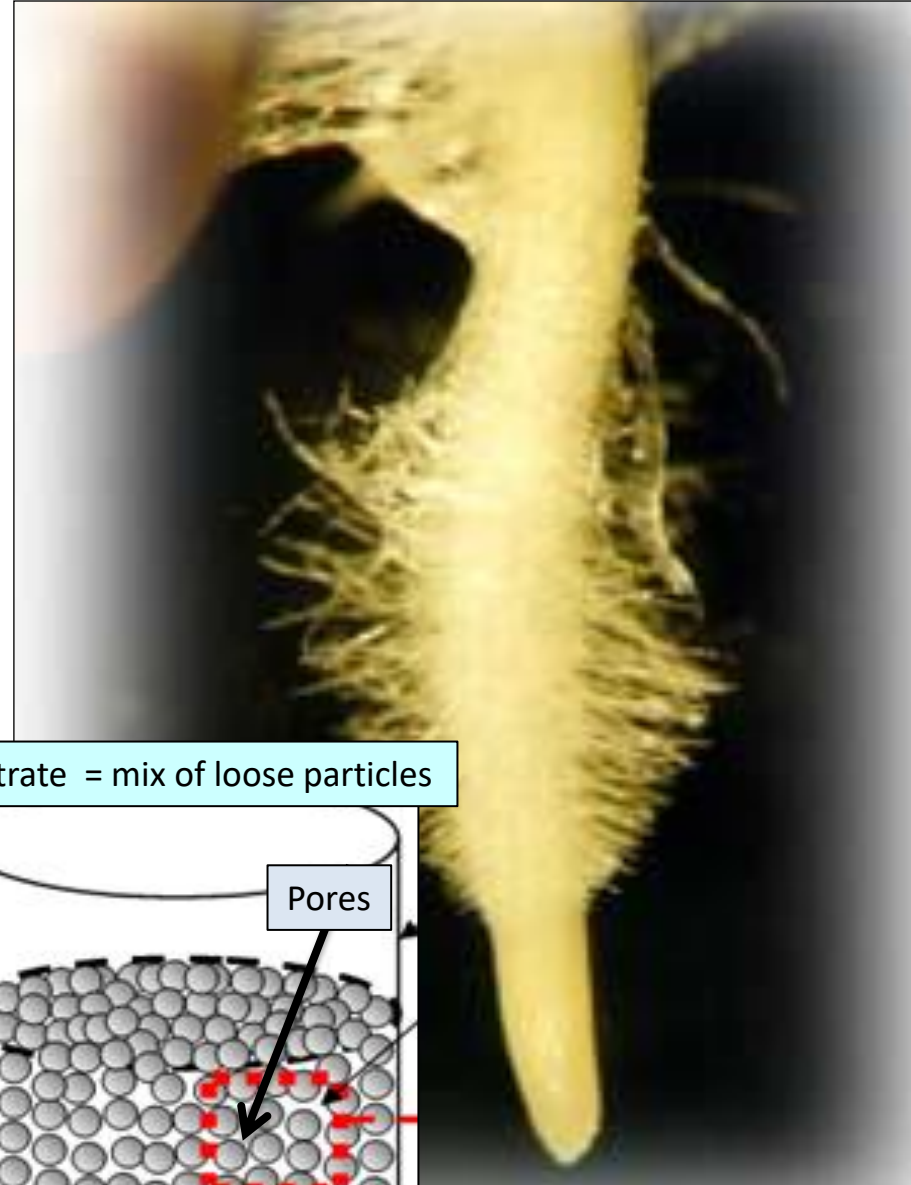
- (projections of epidermis cells).

- **Root hairs** greatly increase the Surface Area over which water can be absorbed.
- **Root hair cells** have a high concentration of ions and dissolved sugars, etc.

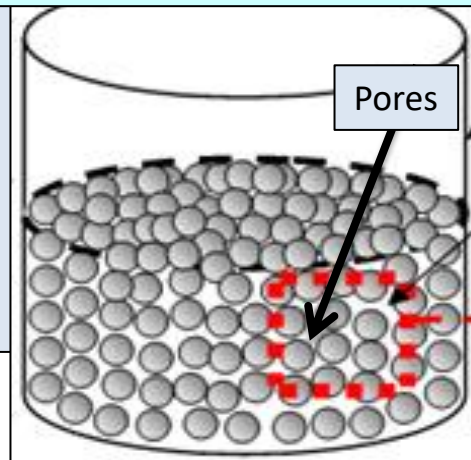
Water enters root hair cells by **Osmosis**

- to even out the Osmotic gradient
- i.e the difference in ionic/solute concentration
- between the inside (cell cytoplasm)
- and the outside (soil solution).

Osmosis = water movement across a semi-permeable membrane.



Substrate = mix of loose particles



Movement of Water Through Plants

Water continues to move by Osmosis through root hairs, into **Xylem** vessels.

Xylem vessels are hollow, semi-rigid, water-proof tubes running throughout the plant.

The diameter of individual xylem vessels is narrow.

- This enables water molecules to creep slowly upwards by **Capillary action**.
- (i.e. cohesion between water molecules
- & adhesion of molecules to side of vessels).
- as in a thermometer.

Osmotic Root pressure & Capillary action prevents downwards water movement in Xylem Vessels

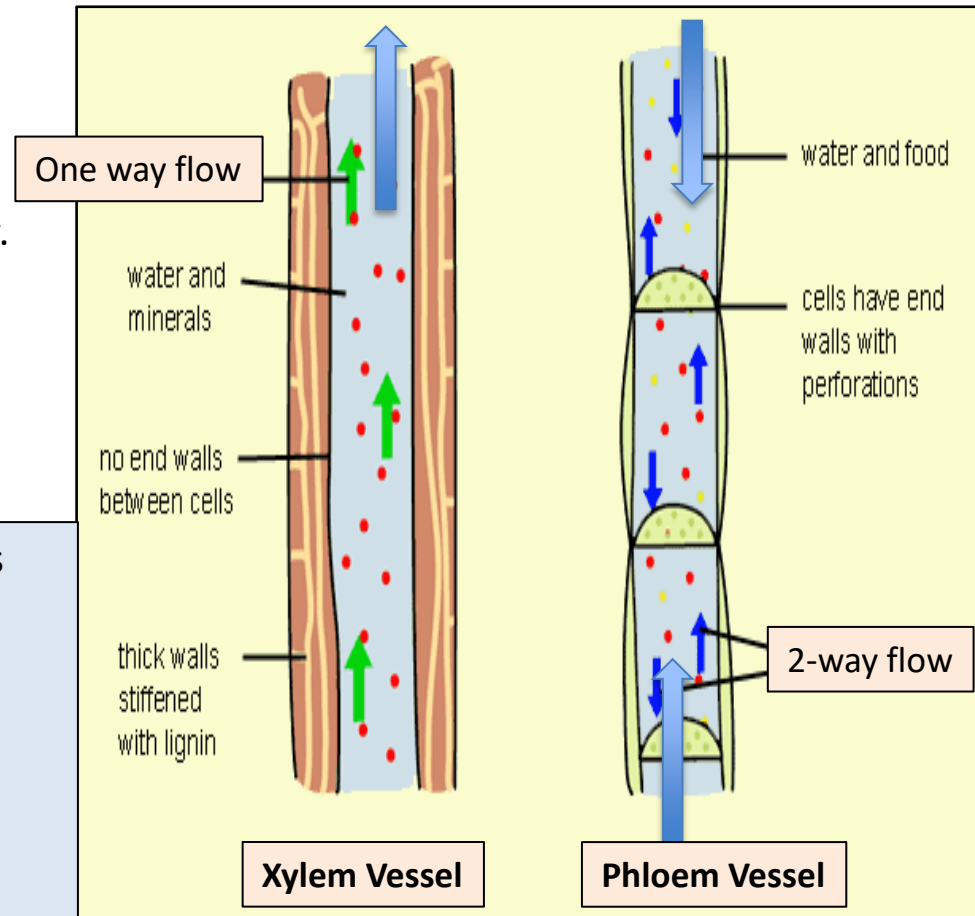
- (One-way flow).

A small amount of water also enters the **Phloem** system

- used mainly to transport sugars & nutrients around the plant.

Phloem vessels have perforated end walls

- Allows Two-way flow
- (from roots to shoots & vice-versa).



Exit of Water from Plants (via Stomata)

Water exits plants, mainly via their stomata

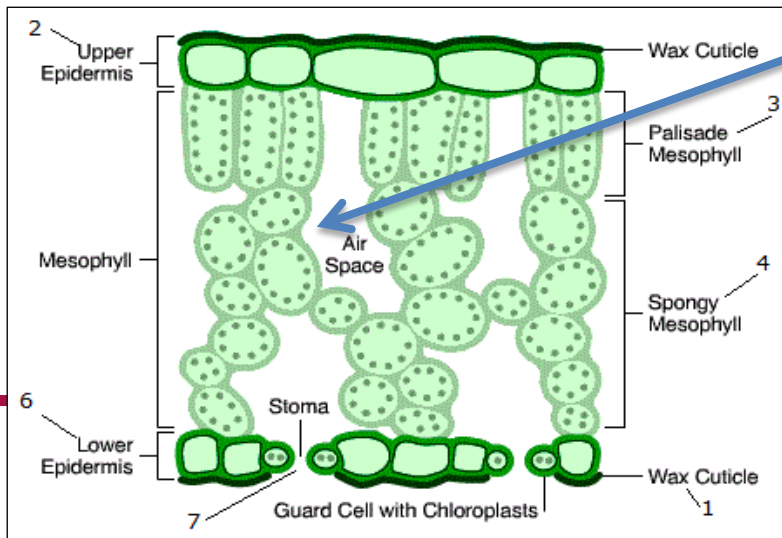
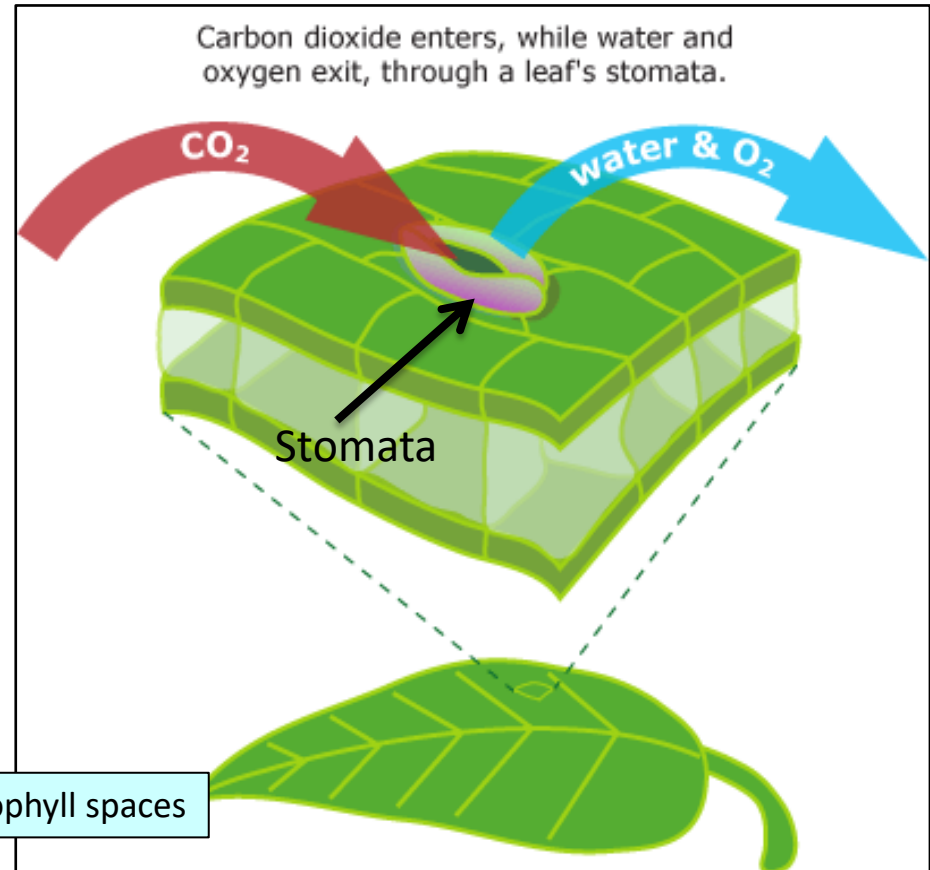
- (pores).
- Most stomata are found in leaves - due to their large surface area.
- Some water can also exit via their waxy cuticle.

Stomata are mainly present on the leaf under-side

- (to mitigate excess moisture loss).
- Water vapour moves from the Xylem vessels
- into Stomata, via spaces in the Mesophyll layer.

Stomata also allow the exchange of Oxygen & CO₂

- Used in Photosynthesis/Respiration.



Mesophyll spaces

Guard cells at the entrance of stomata can open or close

- This regulates the amount of water vapour lost
- and the exchange of CO₂ + Oxygen.

Stressed plant = Closed stomata during the day = **disaster!!**

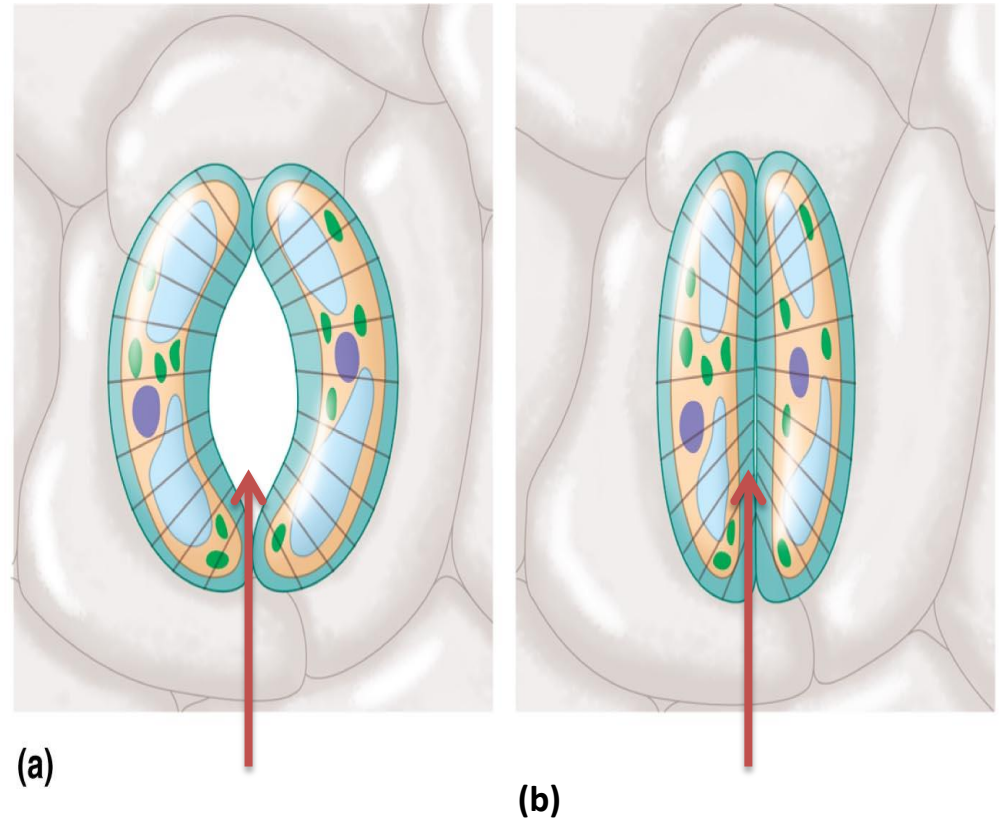
- **No transpiration or Photosynthesis!**
- (Hot leaves, no Ca supply, no Sugars).

Delphy

Regulation of Plant Water Loss by Stomatal Guard Cells

Stomatal Guard cells have uneven thickening of the cell walls

- which allows the aperture to be opened or closed.
- a) When the surrounding cells are well supplied with water (i.e. **Turgid**)
- the guard cells have a higher level of Potassium.
 - Water moves into them by Osmosis, **opening** the aperture.
- b) When the surrounding cells lose water (i.e. **become Flaccid**)
- water moves out of the guard cells
 - causing the aperture **to close**.



Turgour pressure in surrounding cells rises & falls according to the difference between:

- plant water loss (mainly due to evaporation through stomata)
- & water supply from the roots.

What Factors Control Plant Water Use (Transpiration)?

Transpiration accounts for 90% of plant water use.

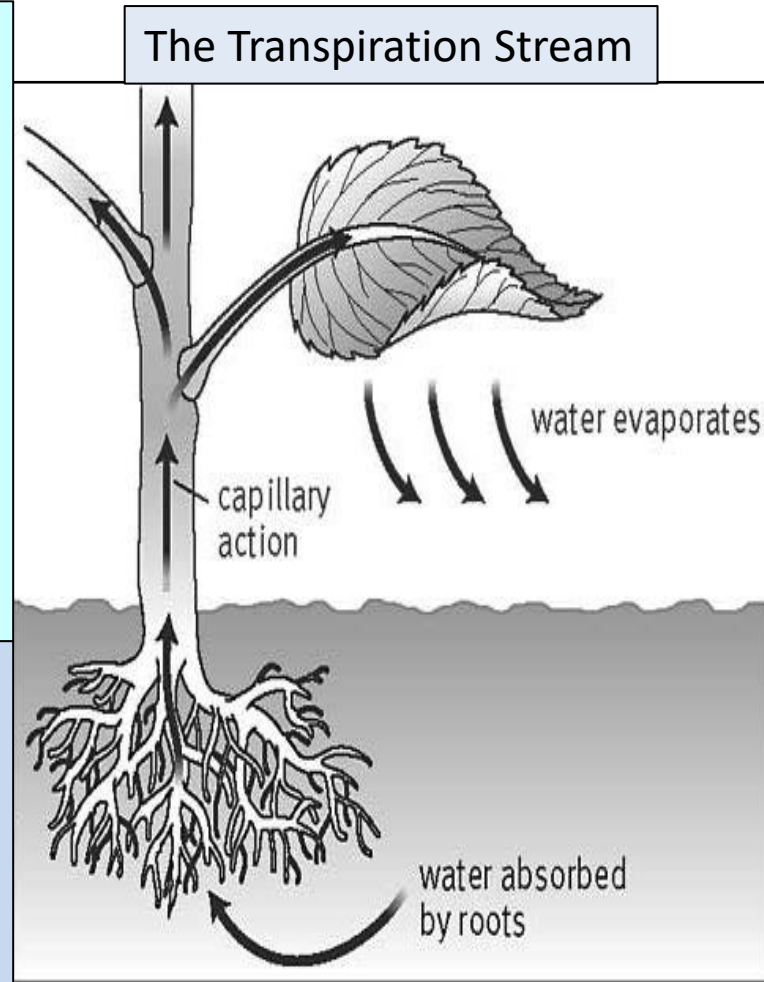
- Uptake and movement of water, via the “**Transpiration Stream**” is controlled in 2 ways:

Water Uptake (Push) Factors

- Osmotic entry of water from pores in the substrate**
 - into **Plant Root Hair Cells**.
 - There is always a solute concentration difference between the pore water & root hair cytoplasm.
 - Water molecules move from pores across the root hair cell walls (membranes) to even out the concentration difference.
 - Osmosis happens night and day
 - (It's a Physical not a Metabolic Process).
- Capillary Forces** then assist water rise within the Xylem.

Water Loss (Pull) Factors

- Evaporation of Water from Open Stomata**
 - (a climate controlled process).
 - Growing Conditions which promote good evaporation increase water loss.
- Aperture of Stomatal Opening**
 - Guard cells** at the entrance of stomata open & close according to the stress experienced by the plant.
 - (Change in water content of the leaf mesophyll layer).



What factors Act on Water Uptake in Substrate Grown Plants?

1. Root Volume & Health

- Larger root volume = larger surface area for water (+ nutrient) uptake.
Poor rooting (or root damage from pest/disease/physical disturbance)
- reduces plant water uptake.

2. Substrate Moisture Tension (Matric Potential).

Moisture Tension = the ease with which water is released

- from pores in the substrate
- (for drainage, capillary movement, or root uptake by osmosis).

Tension is expressed as a negative number & is a function of:

- **Substrate Moisture content**
- **Pore size distribution**
- (Smaller pores = hold water more strongly)

The ideal tension in substrate is -10 to -40 mb (-1 to -4 kPa).

- This is known as **Easily Available Water**.

Pore size in substrate is closely correlated with:

- particle size distribution, root ingress & compaction.

As any substrate ages, its pore size distribution changes:

- (due to particle decomposition/root ingress & compaction).
- Older substrate holds more water - but at higher tension.

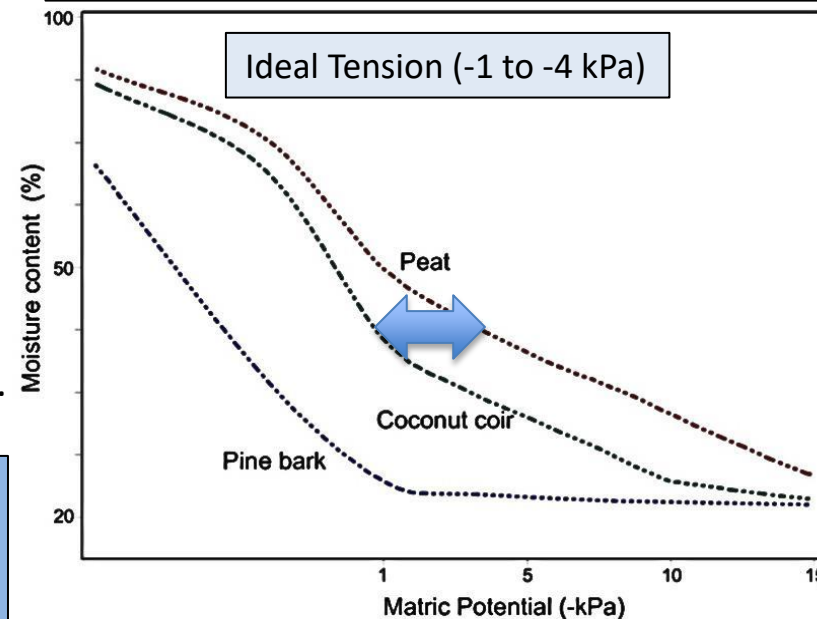
3. Root Zone EC

High Root Zone EC restricts plant water uptake:

- By reducing the difference in solute concentration
 - between the Pore Water and Root Hair Cytoplasm.
- Reduces Osmotic Gradient between inside & outside of root.



Moisture Release Curves for 3 different substrates



What Factors Act on Evaporation of Water from Leaves?

1. Plant Type

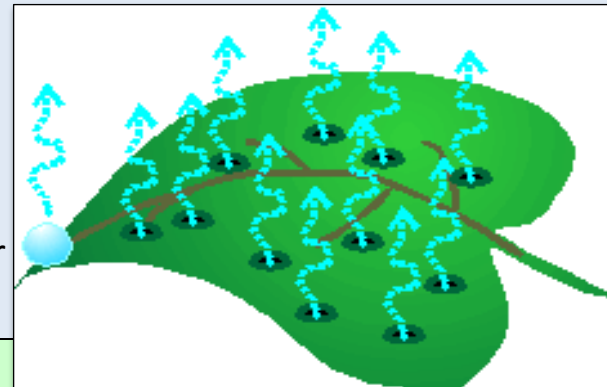
Some plants (e.g. Cacti/Succulents) are adapted to retain water and lose v little in Transpiration.

- **Berry crops** are adapted to transpire freely for optimum yield.

2. Leaf Area

Larger plants with more leaf canopy have more stomata.

- So lose more water in Transpiration (higher water demand).
- Sudden loss of leaf (e.g. cutback) massively reduces stomata number
- Easy to over-water (due to sudden reduction in water demand).



3. Air Temperature

Warmer air increases evaporation from open stomata.

- For field grown (unheated) crops, air temperature is closely correlated with Solar Radiation.
- Under conditions of high Solar Radiation (high air temps), stomata may close to conserve water.
- Growing under shade net (or diffusing/reflecting poly) limits temperature rise & crop water use/stress.

4. Relative Humidity (Moisture content of Air)

If the air is very humid (e.g. $RH = \geq 85\%$), less evaporation of water occurs from stomata

- As there is less difference between saturated air inside the stomata & outside air (less water potential)

If the air is very dry ($RH \leq 45\%$), water loss may be too fast, causing stomatal closure (to conserve water).

In Summer, the ideal Relative Humidity range Berries is around 60-80% - see later slide on VPD.

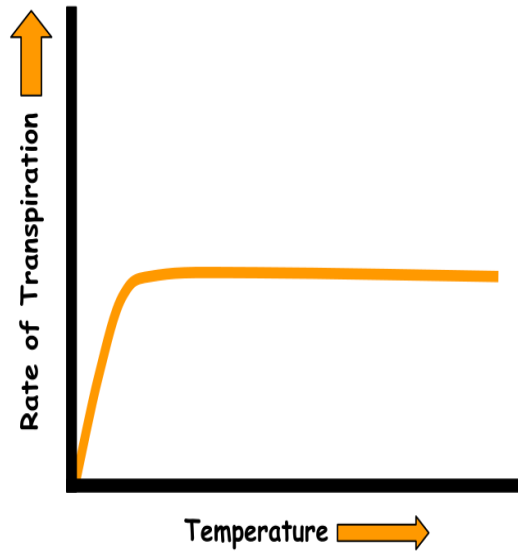
5. Wind Speed

Gentle air movement over leaves, reduces the “**boundary layer**” of still humid area air round stomata

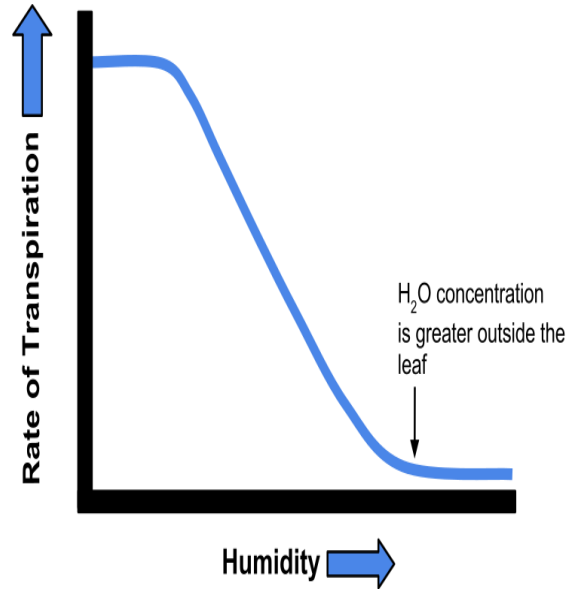
- (Increases water loss).
- Even at high RH ($>90\%$), gentle air movement can stimulate transpiration (Boundary Layer $RH = \geq 95\%$).
- Increasing wind speed ($>3 \text{ m sec}^{-1}$) causes some Stomatal closure - to reduce excess water loss.

Summary - Effects of Temperature, Humidity & Wind Velocity on Transpiration.

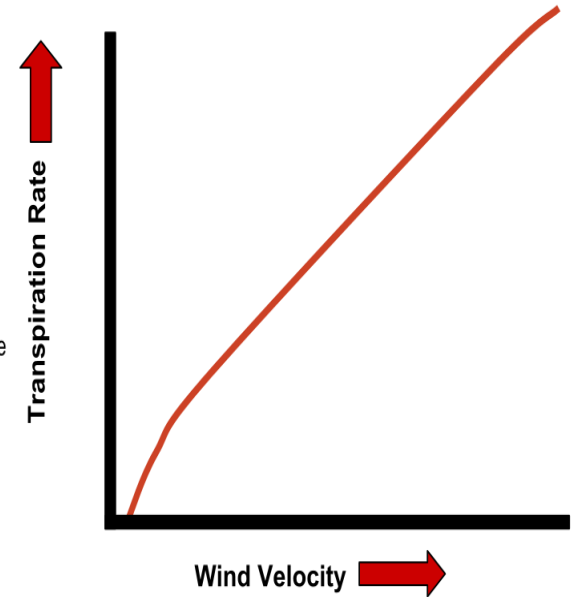
Effect of Temperature on Plant Transpiration



Effect of Humidity on Plant Transpiration



Effect of Wind Velocity on Plant Transpiration



Temperature – Increasing Temperature increases evaporation; until Stomata close.

Humidity – Decreasing Humidity increases evaporation; until Stomata close.

Wind Velocity – Increasing wind Speed increases evaporation; even when Stomata close.

- Very high wind speeds continue to desiccate leaves - due to cuticular water loss.

Note: As stomata are also used for taking in CO₂ for Photosynthesis.

A stressed plant (i.e. one with closed stomata during the day) = **bad news!**

- **No transpiration or Photosynthesis!**

- (Hot leaves, no Ca supply, no Gas Exchange – all bad for optimal yield!

Even **after** stress events subside; **Lag phase** of several hrs before stomata re-open.

Short term stress events have a longer term impact on Crop Growth + Development.

Vapour Pressure Deficit (VPD)

From an evaporation perspective, the effect of air temperature & humidity are closely inter-twined.
Transpiration potential at any given Temperature & Humidity level is termed the Vapour Pressure Deficit

Vapour Pressure Deficit (VPD) is defined as:

The difference (deficit) between :

- the pressure exerted by water vapour in ambient air
- and the pressure exerted by water vapour in saturated air (100% RH).

As the Humidity of air inside the plant stomata is generally 100%:

- Transpiration (VPD) increases with increasingly warmer & drier air
- Transpiration (VPD) decreases with Increasingly cooler & more humid air.

VPD is generally expressed in pressure units (similar to Tension):

- Either in kilo-Pascals (kPa)
- or in milli-bars (mb)
- (where 10mb = 1 kPa).

VAPOR PRESSURE DEFICIT CHART BY: FREGROWL!

TEMP		RELATIVE HUMIDITY													
C	F	100%	95%	90%	85%	80%	75%	70%	65%	60%	55%	50%	45%	40%	35%
15	59	0.0	0.8	1.7	2.5	3.4	4.2	5.1	5.9	6.8	7.6	8.5	9.4	10.2	11.1
16	61	0.0	0.9	1.8	2.8	3.7	4.6	5.5	6.4	7.3	8.2	9.1	10.0	10.9	11.8
17	63	0.0	1.0	2.0	2.9	3.9	4.9	5.8	6.8	7.8	8.8	9.7	10.6	11.6	12.6
18	64	0.0	1.0	2.0	3.1	4.1	5.0	6.2	7.2	8.2	9.3	10.3	11.3	12.4	13.4
19	66	0.0	1.1	2.2	3.3	4.4	5.5	6.6	7.7	8.8	9.9	11.0	12.1	13.2	14.3
20	68	0.0	1.2	2.4	3.5	4.7	5.9	7.0	8.2	9.4	10.6	11.7	12.8	14.0	15.2
21	70	0.0	1.2	2.4	3.7	4.9	6.2	7.4	8.6	9.9	11.1	12.4	13.7	14.9	16.1
22	72	0.0	1.3	2.6	3.9	5.3	6.6	7.9	9.2	10.5	11.9	13.2	14.5	15.8	17.2
23	73	0.0	1.4	2.8	4.2	5.6	7.0	8.5	9.9	11.3	12.7	14.1	15.4	16.8	18.2
24	75	0.0	1.5	3.0	4.5	5.9	7.4	8.9	10.4	11.9	13.4	14.9	16.4	17.9	19.4
25	77	0.0	1.6	3.2	4.8	6.4	8.0	9.5	11.1	12.7	14.3	15.9	17.4	19.0	20.5
26	79	0.0	1.7	3.4	5.1	6.7	8.4	10.1	11.8	13.4	15.1	16.8	18.4	20.1	21.8
27	81	0.0	1.8	3.5	5.3	7.1	8.9	10.7	12.4	14.3	16.0	17.8	19.6	21.3	23.1
28	82	0.0	1.9	3.8	5.7	7.6	9.5	11.4	13.3	15.1	17.0	18.9	20.7	22.6	24.5
29	84	0.0	2.0	4.0	6.0	8.0	10.0	12.0	14.0	16.0	18.0	20.0	22.1	24.1	26.1
30	86	0.0	2.1	4.2	6.4	8.5	10.6	12.7	14.8	17.0	19.1	21.2	23.3	25.4	27.5
31	88	0.0	2.2	4.5	6.7	9.0	11.2	13.4	15.7	17.9	21.2	22.4	24.6	26.9	29.1
32	90	0.0	2.4	4.7	7.1	9.5	11.9	14.2	16.6	19.0	21.3	23.7	26.1	28.4	30.8
33	91	0.0	2.5	5.0	7.5	10.0	12.5	15.0	17.6	20.1	22.6	25.1	27.6	30.1	32.6
34	93	0.0	2.7	5.3	8.0	10.6	13.3	15.9	18.6	21.2	23.9	26.5	29.2	31.8	34.5

Ideal VPD for most plants = 0.75-1.05 kPa (7.5-10.5 mb)

- Green area on chart (Happy plant – freely transpiring).
- As the air temperature increases, it can hold more moisture
- Green area shifts to the left (less vapour pressure at high RH).

If VPD >1.2 kPa (>12 mb)

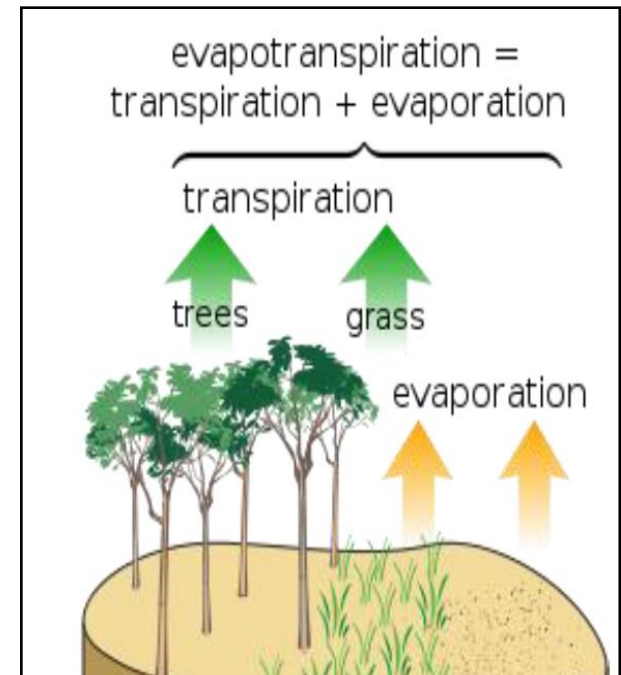
- increasingly higher transpiration rate (onset of stress)

If VPD < 0.5 kPa (<5 mb)

- very low transpiration rate (excess humidity/high disease risk).

Whole Crop Water Use - Evapo-Transpiration (ET)

- ✦ The term **Evapo-Transpiration (ET)** is often used to denote whole crop water use in a field situation.
- ✦ For plants growing in open soil (e.g. Grass, Trees)
 - a significant amount of water is lost by evaporation from the surface layer
- ✦ There is no easy way to distinguish:
 - Evaporation losses from the soil surface
 - and Transpiration from the crop.
- ✦ So, whole crop water use is generally expressed as E-T.
- ✦ For crops grown in bags, pots or troughs:
 - the surface area for evaporation is greatly reduced.
 - especially if the surface of the pots or troughs is covered with mulch (e.g. sawdust, bark/coir mat).
- ✦ Therefore in substrate, evaporative losses from the rooting area of the crop are relatively small.
 - ET is more closely correlated with Transpiration.



Section 2

Substrate Physical Properties

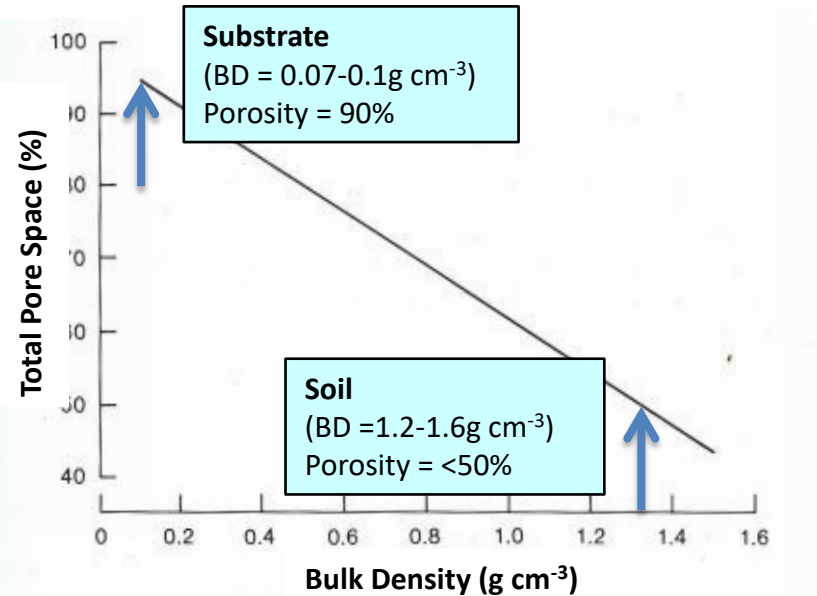


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Why Don't we use Soil for Containerised Production?

The ideal substrate for containerised production has to fulfil major functions:

1. It must be **light-weight** to make it efficient to transport (not to collapse crop support systems).
 - Typical Bulk Density for substrates is $0.07\text{--}0.1\text{ g cm}^{-3}$
 - Typical Bulk Density for soils is $1\text{--}1.5\text{ g cm}^{-3}$ (10x more).
2. It must have **high porosity** to hold a lot of water & air
 - so it can be irrigated frequently, but can also drain fast.
 - Typical porosity of many substrates is 80-90%
 - Typical porosity of mineral soils is 40-50% (<50% of substrates)

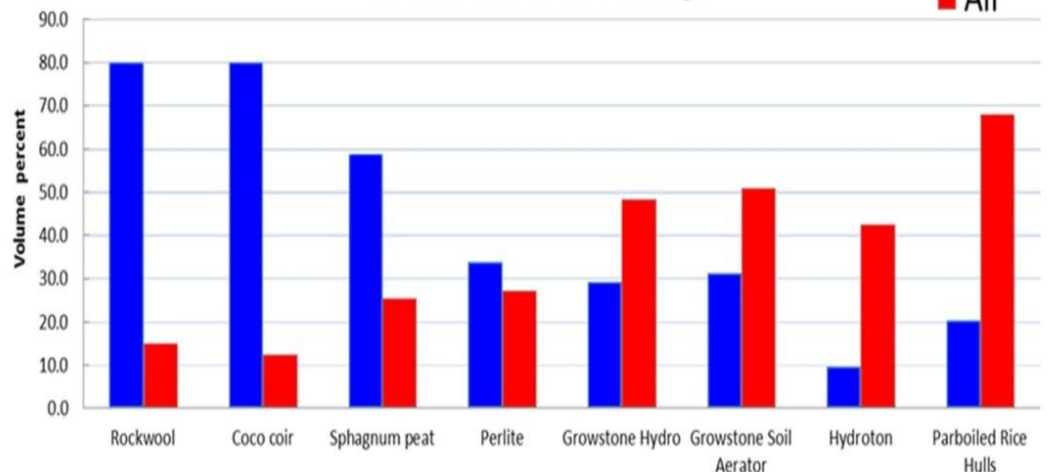


Bulk Density & Pore space are inversely related.

The air & moisture holding balance within a substrate varies:

- according to particle size distribution.
 - Berry substrates should be able to hold:
 - 45-80% of their volume as water*
 - And 10-35% of their volume as air*
- * At Container Capacity (i.e. after Drainage).

Average Water Holding Capacity/ Air-Filled Porosity of Substrates After Drainage



Basic Substrate Physics - Air/Water Relations

Horticultural Substrate consists of many particles of different sizes, all loosely mixed together.

In the spaces between the particles (pores):

- either water or air can be stored.

From a plant perspective:

there are 3 crucial states in air/water balance:

1. Optimal balance (i.e. just below Container Capacity)

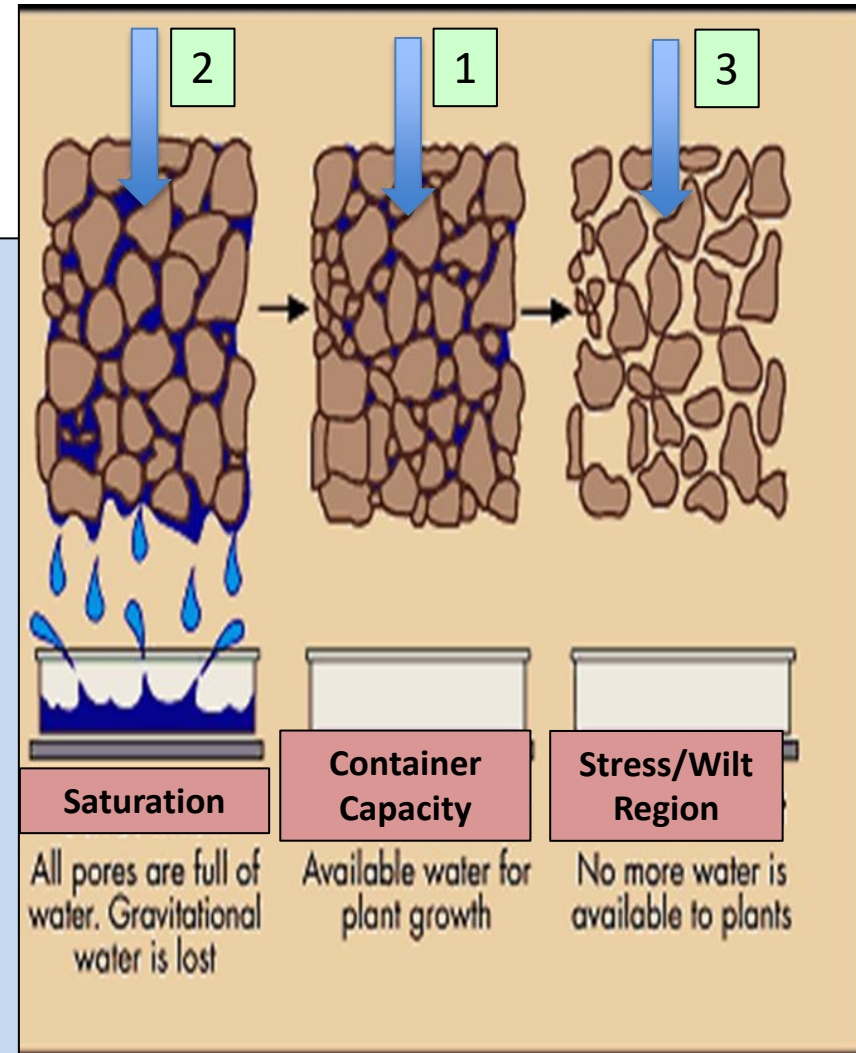
- A narrow band - with easily available water.
- (but sufficient aeration for root health).
- Typically, when Moisture Tension = -10 mb to -40 mb.
- **Container Capacity** follows an irrigation event
- Once excess water has drained by gravity.

2. Substrate Saturation.

- (Typically, when Moisture Tension < -10 mb).
- There is too much water & too little air (oxygen).
- Leads to root death & rapid decline in plant growth.
- Easily occurs if crop over-watered, or drainage is poor.

3. Plant Stress/Wilt region

- (Typically when Moisture Tension > -50 mb)
- There is too much air & insufficient water.
- As substrate dries out, plant growth (& fruit size) is reduced.
- Eventual wilting (at > -100 mb) & plant death (at > -200 mb).



Understanding Matric Potential (Tension) in Substrate

The size of particles that make up the substrate

- affects the size of pores (Pore size distribution).
- Pore size dictates how strongly water is held
- (by the forces of adhesion and cohesion).
- i.e. how easily water is lost from the pores.

Pore size in soil/substrate is classified as follows:

- **Large pores** (+75 μm in diameter)
- lose water very easily (at <-5 kPa tension).
- **Medium pores** (30-75 μm in diameter)
- lose water less easily (at -5 to -10 kPa tension).
- **Micro pores** ($<30\mu\text{m}$ in diameter)
- lose their water only at high tension (>-10 kPa).

In fruiting crops, we aim to optimise plant growth:

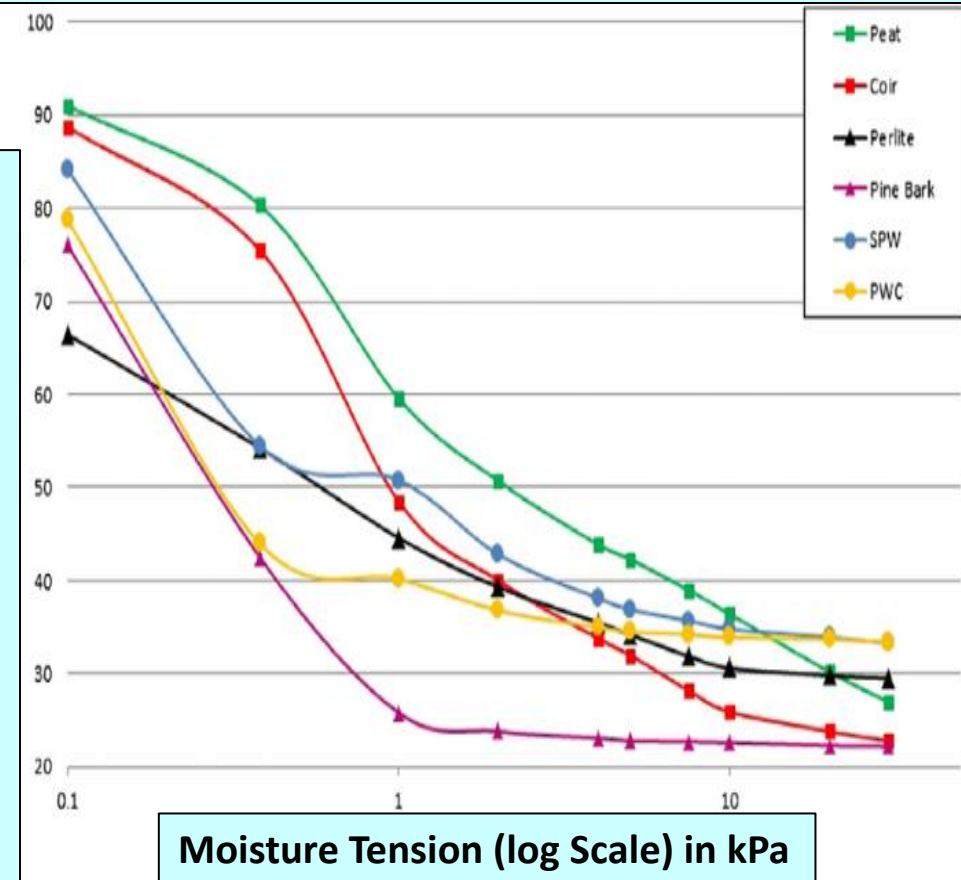
- by keeping Moisture Tension at -1 to -4 kPa.

Only Macro-pores release water at this tension.

However, it is important to have some reserve capacity of water (buffer capacity):

- (i.e. some medium size pores).
- To avoid severe plant damage if
- the substrate dries out & tension rises >-5 kPa.

Example Retention Curves of Different Substrates
(-1 kpa = -10 mbar = 10 cm water tension)



To understand how the particle size distribution affects a substrate's ability to hold water:

- Moisture Retention (pF) curves are useful.

From these, we can relate what the measured moisture content should be at, any given moisture tension.

What is the optimal tension for Substrate Berry Crops?

De Boodt & Verdunek determined that for the optimal moisture tension for peat grown tomatoes was:

- -1 to -5 kPa (defined as easily available water).
- At tensions higher than this, transpiration rates decreased and fruit size in tomatoes declined
- (indicating the onset of moisture stress).

More recent work with other crops (e.g. Roses) grown in Perlite + Coir:

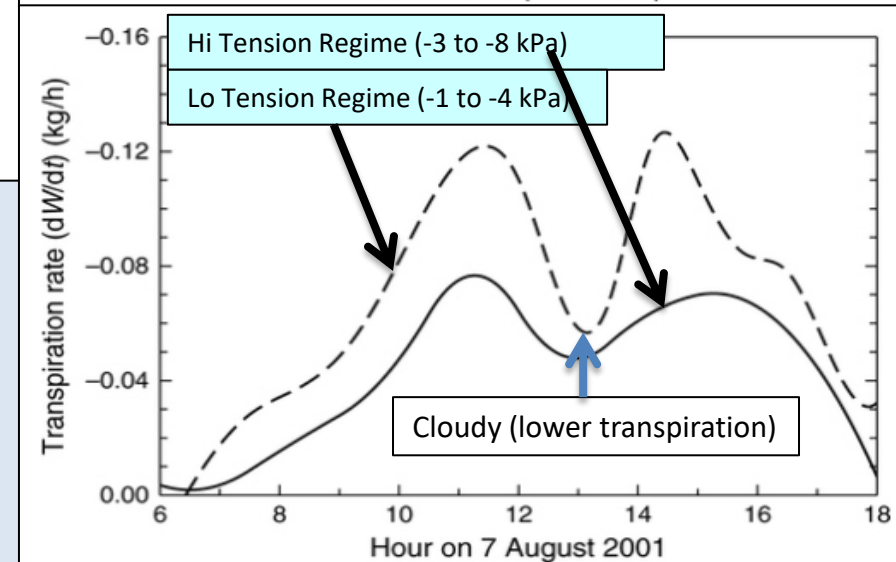
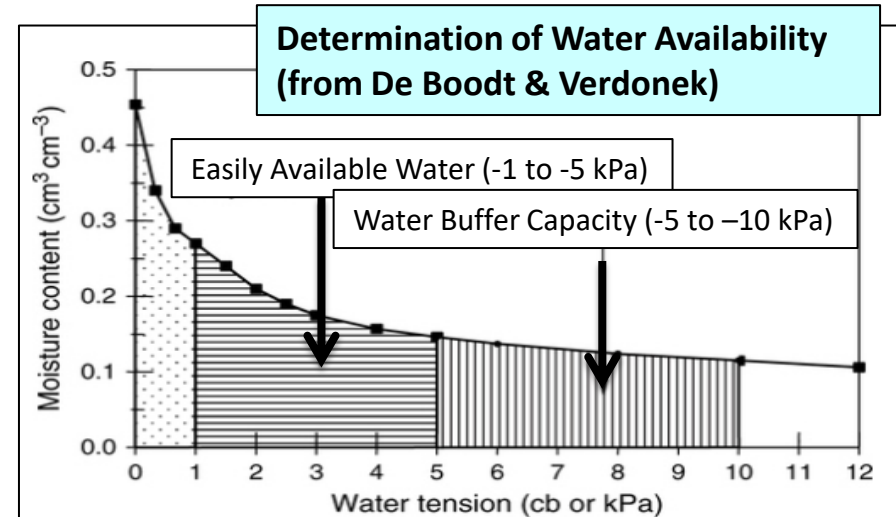
- Showed a similar decline in transpiration rates & yield at tensions above 4 mb.
- Differences in Transpiration rates were less evident under cooler (lower transpiration) conditions

Although there are no detailed studies in berry crops

- Fruiting crops would be at least as sensitive as
- Roses & Tomatoes, to increasing moisture tension.
- A suggested optimal moisture tension range of
- -10 to -40 mb for peat/coir grown berries seems appropriate.
- Under lower transpiration conditions
- slightly higher tensions (-50 to -60mb) may be ok?

For nursery crops, or where growth control is required

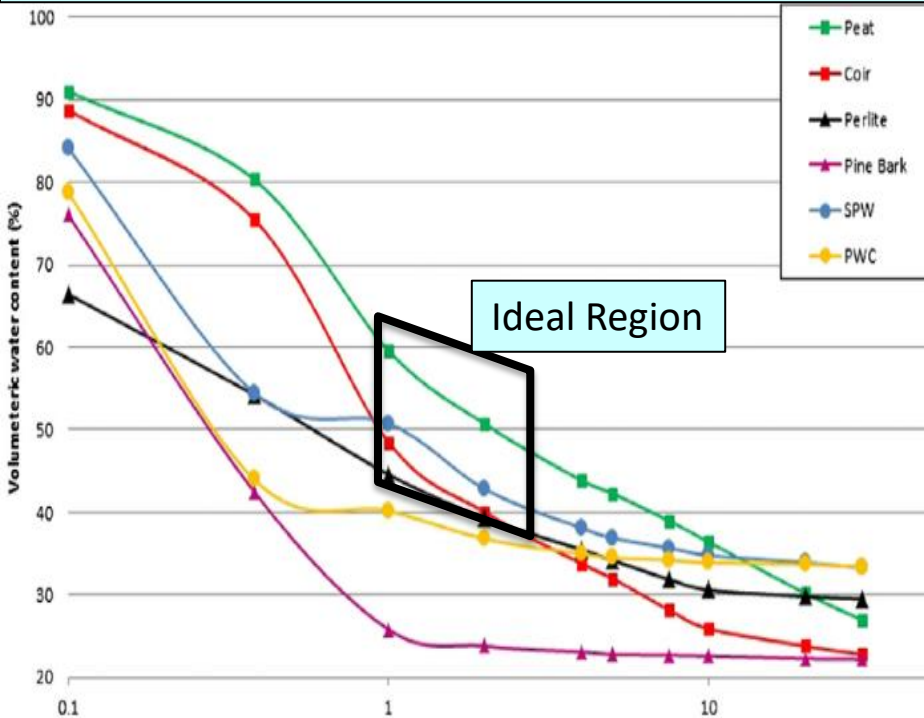
- (e.g. for vigorous primocanes)
- Higher tensions (-50- to 60 mb) are more normal?



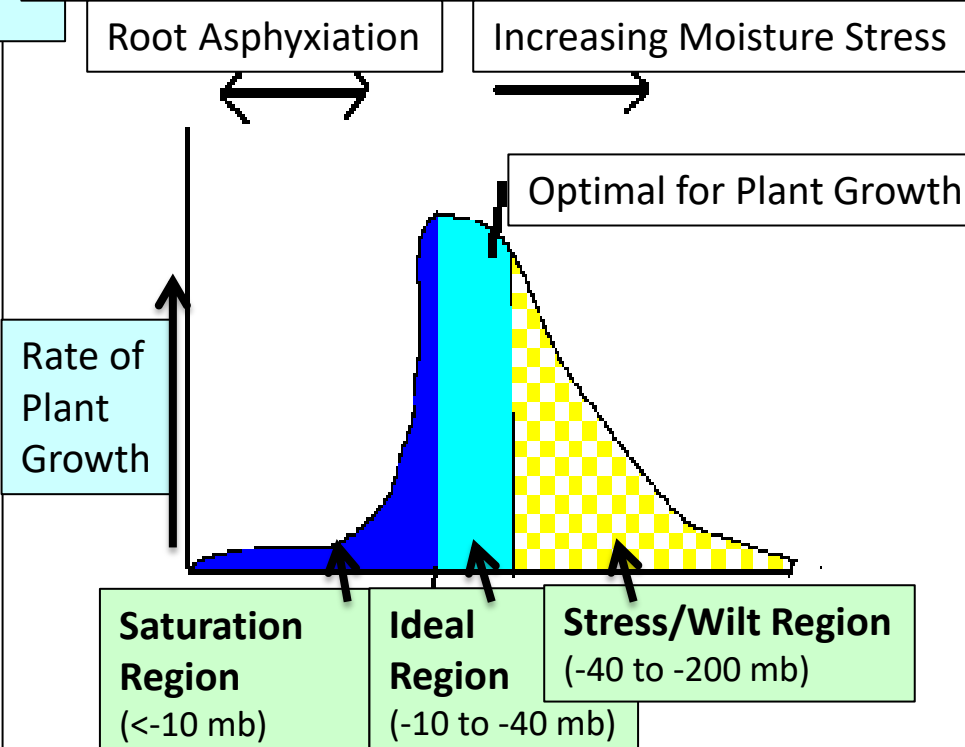
Transpiration Rates of Roses under Low & Higher Tension (Wet and Dry) Irrigation Regimes.

Equating Moisture Tension to Plant Growth & Substrate Moisture Content

Moisture Retention Curves - Various Substrate Mixes
(-1 kpa = -10 mbar = 10 cm water tension)



Plant Growth Curve in Relation to Moisture Tension



- **Peat** (Green curve) = 60% moisture content at -10 mb tension.
- **Coir** (Red curve) = 48% moisture content at -10 mb. (Has fewer fine particles/less small pores).
- **Bark** (Purple curve) = only 25% moisture at -10 mb. Loses water too easily – Too thirsty for most crops! (V large particle size – Too many very large pores).

- The measured % volumetric moisture content of
- any substrate at any tension level:
 - is dependant on its particle size distribution.
- Particle size is more important than raw material**
- An Ideal Berry substrate holds 40-65% moisture
- at a tension of -10 to -40 mb (when new)

Which Substrate for Berry Crops?

The early history of substrate Berry production was dominated by the use of Peat.

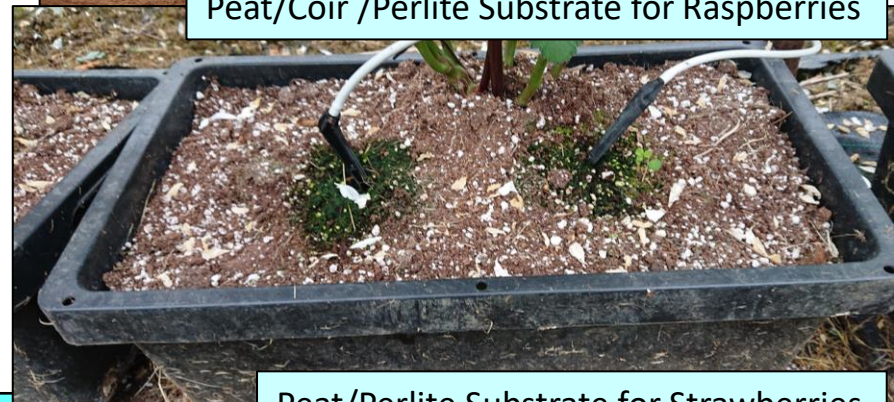
- It was plentiful, relatively cheap + light-weight.
- It can also hold a lot of water (60-75%) at relatively low tension (<-50 mb)
- Substrates with different physical properties
- can be achieved by blending different Peat types
- (e.g. White + Dark, Sod-Cut & surface-milled)
- and by grading it into different size fractions
- More recently, mixes of Peat, Coir and Perlite have become popular (to improve drainage + longevity).

However, due to increasing environmental pressure over unsustainable levels of Peat Extraction

- Peat-free alternatives have been actively sought.



Surface Peat Milling



Peat/Coir /Perlite Substrate for Raspberries

Peat/Perlite Substrate for Strawberries



Delphy

The rise of Coir:

- Over the past 15 years, 100% Coir based substrate has replaced
- much of the peat-based substrate in many berry-growing areas.
- Although chemically more challenging, Coir's physical characteristics
- make it even suited to Berry cultivation.
- It has more medium to large pores, so holds enough available water
- but has more air at lower tensions (<-50 mb).
- (Gives better roots and easier water management).
- Even with Blueberries, Coir-based mixes are gaining in popularity.

Peat Substrate - Particle Size Distribution + AFP

Raw peat consists of particles of many different sizes:

- from <1mm (dust) to >32 mm (coarse).

Peat is processed by passing across different mesh screens:

- to separate it into various different particle size fractions.

A range of substrates can be made from blends of different screen size fractions

- according to specific recipes.

Blends are often classified according to

- Particle size distribution:
- (e.g. Fine, Med/Fine, Med/Coarse).

In many blends, a high proportion of fine & small fractions is used (>40% by volume)

- to help retain water (more small pores).

Coarser (more free-draining) substrates

- have higher % of large/coarse fractions.
- (more large pores = higher AFP).

Particle Size (Fraction)	Medium/Fine Substrate (AFP* = 8.2)	Medium/Coarse Substrate (AFP* = 14.4)
<1 mm (dust)	21.5	18.9
1-2 mm (fine)	13.5	12.5
2-4 mm (small)	10.9	12.9
4-8 mm (medium)	11.9	17.1
8-16 mm (large)	17.2	18.2
16-32 mm (v large)	19.5	2.9
32 mm+ (coarse)	5.4	17.5



Medium/fine fraction (1-8 mm)

AFP = Air Filled Porosity.

This is a measure of the amount of pores filled with air

- once the compost has drained after saturation
- (i.e. at container capacity).

For Strawberries, an AFP of 14-18% is considered ideal.

For Raspberries an AFP of 18-21% is more desirable.

Making Coir Substrate for Berry Crops

Coir Substrate for Berry Crops is made from a mixture of 2

- (or sometimes 3) different raw materials.
- All derived from the husk of harvested coconuts.

Short Fibre. Extracted from aged (brown) coconut husks.

- Generally, fibres of 20-35mm length & 1 mm diameter.
- Used in varying amounts (For Berries, typically 10%-35%)
- Mainly used to improve lateral movement of water
- Also reduces Water Holding Capacity.

Pith. Dust left-over from the fibre extraction process.

- Consists mainly of particles of 0.5mm to 5 mm diameter.
- Has a high water & air holding capacity + easily re-wets.
- For Berries, typically makes up 50-90% of the final product.

Chips. Made by milling (chopping) husks into chunks

- of varying diameter (typically 5-15 mm).
- Larger particles = more large pores, so less water retention
- Typically used as a 15-35% incorporation.
- Gives some longer term structural stability (over 3-5 yrs).

Ultra Chips.

Milled Green Husks of smaller (2-5 mm) size.

- Used by some producers instead of larger chips
- to improve longevity but with better water retention.
- But, less long term structural stability (2-3 years?)

Short Fibre



Coir Pith



Ultra Chips



Coir Chips

Typical Physical make-up of Coir Substrate (Coco-Peat)

Coir pith comprises mainly of small particles (0.5- <4 mm).

- Unprocessed Pith has a high content of fine dust (0.5-1 mm)
- Holds up to 90% moisture at Container Capacity
- due to a large amount of v small pores.

For Berry substrate, much of the 0.5-1 mm fraction should be removed:

- This increases aeration, by increasing the average pore size.

Some short fibre (10-15%) is also present

- which improves lateral movement of water.

Typical Strawberry Coco-peat mixes

- (containing 85% Pith and 15% Fibre)
- initially hold 55% moisture at -25 mb tension.

Extra fibre, further reduces water holding capacity.

- Up to 30% Fibre may be added to mixes requiring less water retention (45% at -25 mb).
- (e.g. Winter Raspberry Mixes).

Particle Size Distribution of Typical Coco-Peat

Particle Size Distribution	Average
0,5-1 mm	10 %
1-2 mm	20 %
2-3 mm	35 %
3-4 mm	25 %
Fiber	10 %



Close Up of 85:15 (Pith:Fibre) Strawberry Substrate

- Screened to remove all particles <1 mm

Adjusting Air/Water Holding Capacity in Coir Substrate.

The **air/water balance** of Coir Substrate can also be altered by changing particle size distribution.

Unprocessed Pith contains many fine particles (<1mm diameter)

- Retains >80% of its volume as water (at -10 mb tension).

If all the particles below 1mm are removed by sieving:

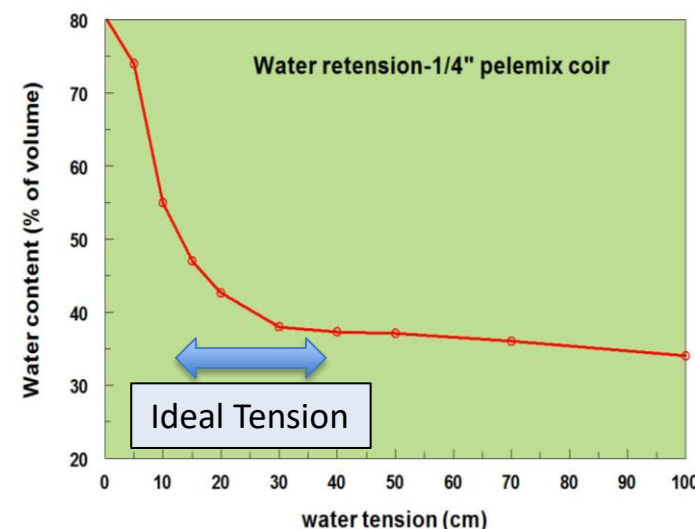
- moisture retention (at -10 mb tension) drops to 65%.

If all the particles below 2mm are removed by sieving

- moisture retention (at -10 mb tension) drops to 40%.

Particle Size of Coir Pith	Water Holding Capacity	Air Filled Space
Small (<1 mm) →	80%	10%
Medium (1-2 mm) →	65%	25%
Large (2-4 mm) →	40%	50%

Water retention - 1.8 mm Sieved Coir Substrate



Many Coir Substrate suppliers offer screened (sieved) product:

- When most of the particles <1mm (or <2 mm) mm removed:
- It gives the substrate a longer working life (over 2-3 seasons)
- But, better moisture retention than by adding extra fibre.

Screened Coir substrates are ideal for Raspberries:

- Typically 1-4 mm Coir Pith with 10-15% added fibre
- Has more larger pores, than unscreened product
- So is less prone to over-watering (Better root development).
- (Less lateral movement than unsieved product with 25% fibre.
- (Needs 2-3 drippers per pot for even pot moisture distribution.

Addition of Chips to Berry Substrate

Adding Chips also reduces moisture holding capacity - due to their larger particle size (5-15 mm).

- Chips decompose more slowly than Pith, so are a good addition for longer term crops (2-3 yrs)
- But, larger pore size = more "thirsty" substrate. Needs more frequent drip cycles to keep above 40% moisture.

Adding Perlite to Substrate Mixes

Perlite is an expanded volcanic material, forming white aggregates (particles).

Horticultural Grade Perlite used in Substrate production typically has particle sizes of 2-5 mm.

Water is held in held pore spaces between the aggregates (not within them).

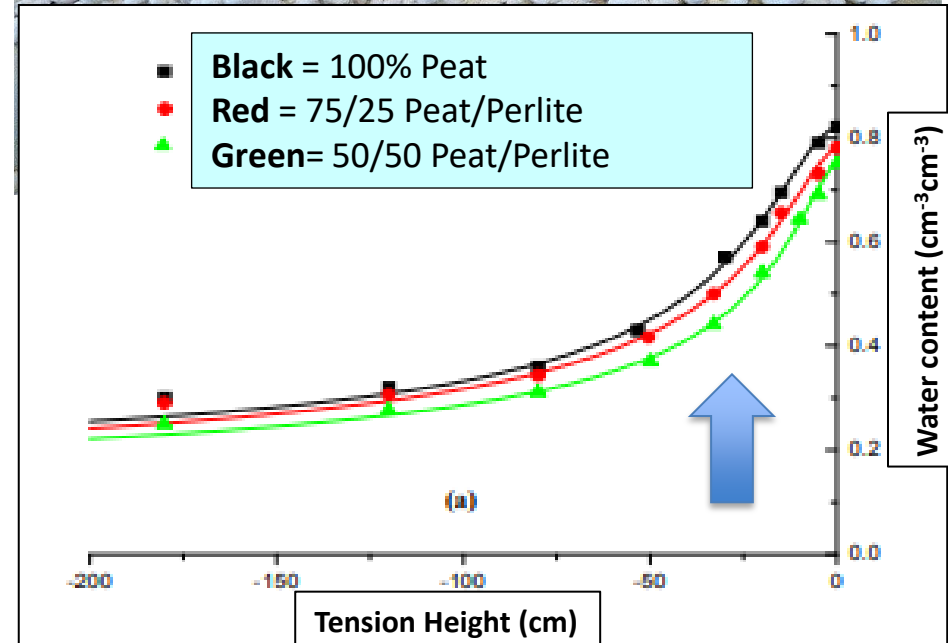
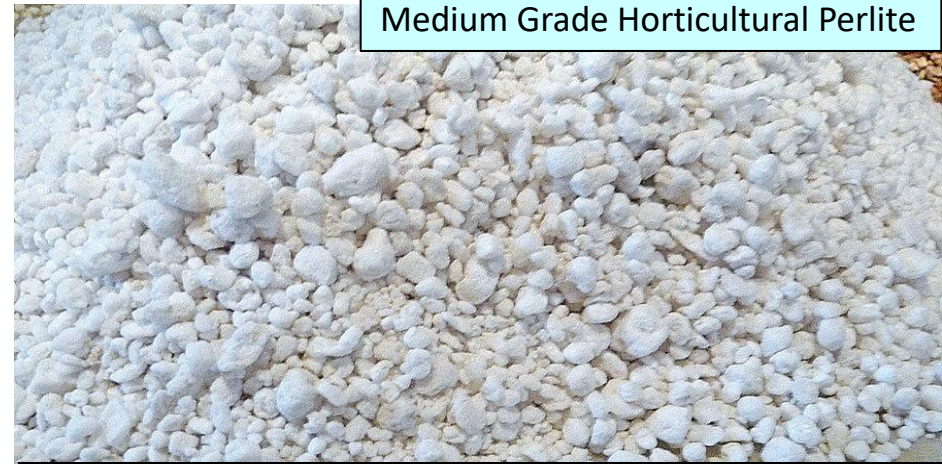
- So pore sizes are large, and drainage is fast.

Adding Perlite to many substrates, reduces water holding capacity

- especially at lower water tensions <-25 mb
- (<-25 cm - see pF curve).

Unlike organic substrate materials (e.g. Peat/Coir):

- Perlite aggregates are stable & do not decompose over time (as with Coir Chips).
- Gives more stability on long term crops?
- (Provided there are not many fines).
- Perlite also has little or no CEC:
- So can help with EC management of substrate



Adding 10-25% Perlite to Peat/Coir substrates with a high % of fine material

- Helps help improve drainage in yr 1; but has less impact as the substrate ages.

Is this the right solution? Maybe better long term to remove fine particles.

Section 3

Assessing Substrate Moisture Content and Run Off



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Assessing Moisture Tension in Substrate Grown Crops

A simple way to assess moisture tension is to carry out a squeeze test.

Take a handful of moist substrate and squeeze to assess how easily moisture appears:

- **Hard Squeeze** with little or no moisture = too dry.
- **Medium Squeeze** to give moisture = little bit dry.
- **Gentle Squeeze** to give moisture = close to Container Capacity (ideal).
- **No need to squeeze** (water already visible) = too wet!



By using a static electronic Micro-tensiometer:

It is possible to measure & record Moisture Tension. This gives the best assessment of plant water availability

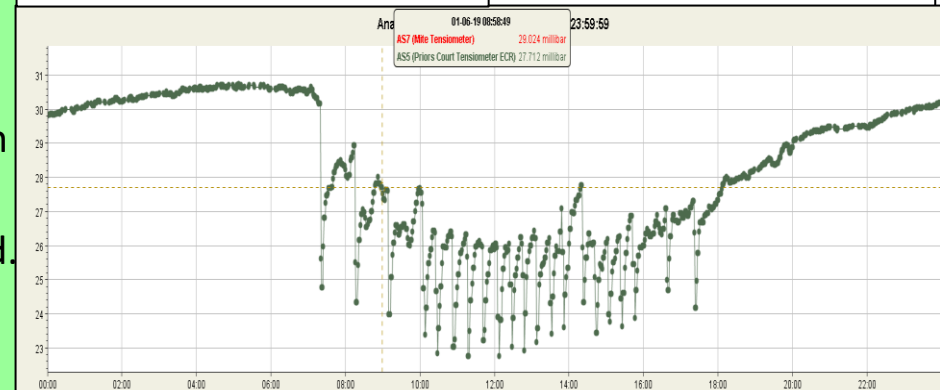
- In different substrates
- And also as the substrate ages:
- Best control of plant water requirement & irrigation scheduling.

However, this technology is currently not widely utilized.

- It is more common to assess the % volumetric moisture content of the substrate
- & base irrigation decisions on this.

Static Electronic Micro-Tensiometer in pot

Medium/Hard Squeeze – bit dry?



Micro-Tensiometer Irrigation Record

Electronic Measurement of Root Zone Moisture in Substrate

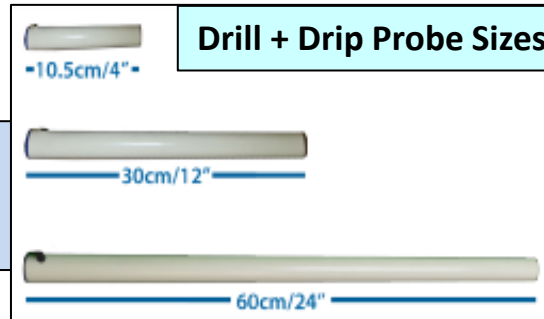
There are two common methods of electronically measuring substrate moisture content:

1. Capacitance Probes (e.g. Sentek Drill & Drop)

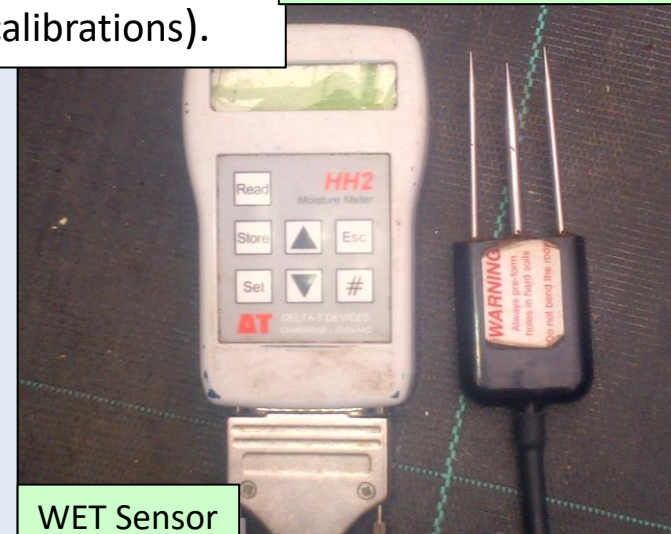
- These measure volumetric soil moisture content every 10 cm, along the length of the probe.
- Various probe lengths available, the most common in substrate being 30 & 60 cm.
- Also, Single Point (10 cm) probes are becoming available – less obtrusive.
- They are expensive, which limits the number of probes that can be employed.
- Their accuracy in substrate is debated too (they have no specific calibrations).

(e.g. Delta T- WET Sensor, Spectrum Field Scout)

- These are probes that generally have 2 or 3 steel rods that can be manually inserted or statically employed.
- TDR probes give an average reading over the length of the rods.
- So, knowledge of the moisture profile is required.
- TDR Probes are said to be more accurate & cheaper than Capacitance Probes.
- Also, they can be employed as roving devices, as well as in static locations.



Drill + Drop in 7 litre pot



WET Sensor

Static vs. Roving Moisture Measurement in substrate

1. Static Devices

Static Devices give real time information on substrate moisture content (and plant water use).

- This helps greatly in improving irrigation scheduling.
- The siting and number of measurement sites used is critical for success.
- They must be representative of the crop being monitored.
- They must also be installed with good contact between probes & substrate, to ensure accuracy.

Due to variations in climate, dripper output, plant size:

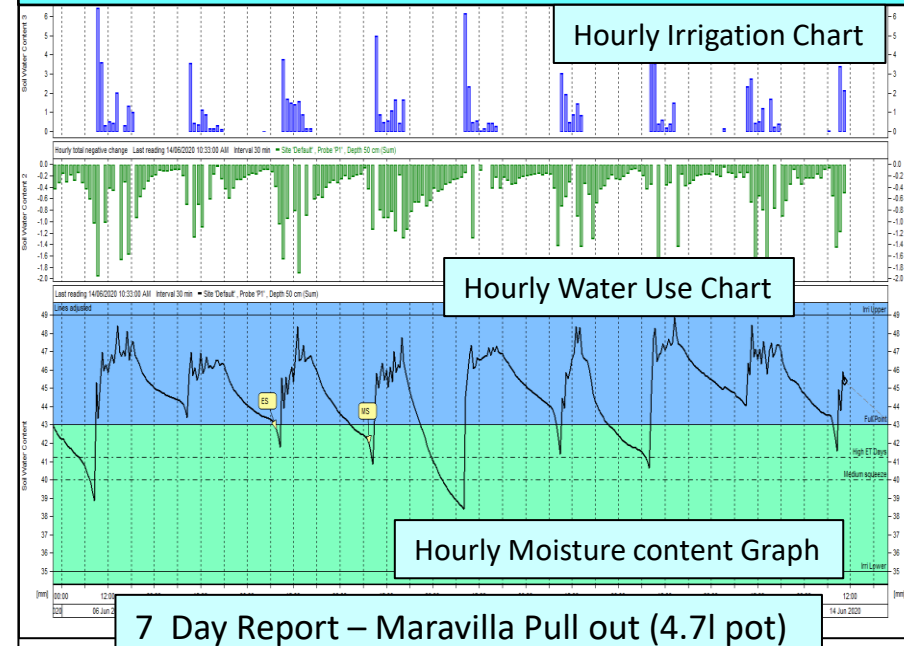
- multiple locations give less erroneous data
- Than from a single device.
- (Ideally use 5 probes per site).

2. Roving Devices

Roving devices don't give real time info

- but allow 100's of readings from different locations .
- This helps take into account the moisture variability
- between individual pots & across the site as a whole.
- You can also take multiple readings from a single pot:
- (Check Moisture levels between & below drippers)
 - to track the horizontal & vertical movement of water
 - and map the moisture variation within.

Graphic Display of Irrigation Inputs & Substrate Moisture Content from Static (Capacitance) Probe



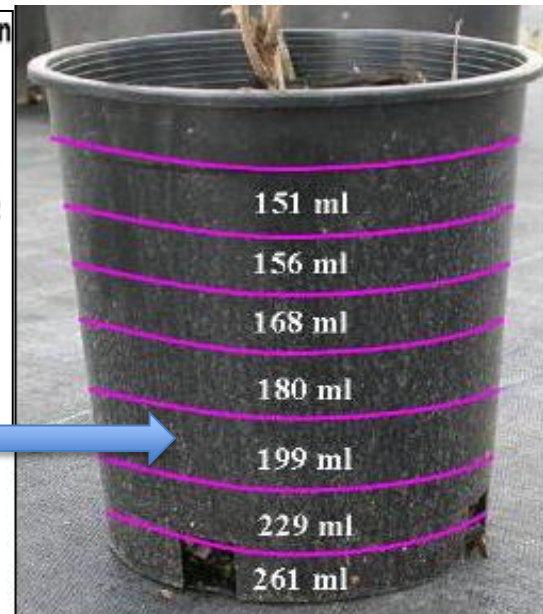
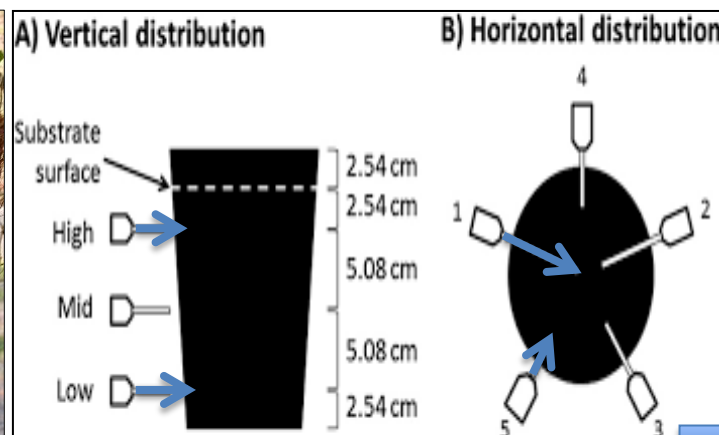
Multi-Sensor Info & Substrate Calibration

TDR Probes can measure EC & Temperature too.

- The WET Sensor can also be purchased with pre-installed calibration data.
- For different substrates (Peat/Coir/Rockwool) Few other TDR (or Capacitance) sensors have substrate calibrations.
- More care is required to interpret readings.

Correct Use of WET Sensor in Pot Grown Crops

- Substrate Moisture content varies according to height of pot and distance from dripper(s).



- Take lots of readings (10-20) per valve, to get decent average.
- Beware of large variations in moisture at different points in irrigation cycle.
- Readings can be stored on HH2 (handset)
- Can export to Excel
- to calculate plot ave moisture & spread of readings.

Variation in Pot Moisture content

- Vertical Gradient** (Gravity)
 - Surface is driest
 - Basal layers are wettest
- Horizontal Gradient** (Capillarity)
 - Immediately around/below the emitter is wettest.
 - Between drippers/furthest point from emitter are driest.

Best **average** - take readings at one third of height of pot (from base)

- Forks oriented **horizontally** (to keep in same horizontal layer).

Guideline Moisture Levels in Fresh Coir Substrate & after 1 Growing Season (Using WET Sensor)

Coir Substrate Type	Composition	Moisture Level in Fresh Substrate	Moisture Level After 1 Season*
Fine (Legro – Air-line)	80% Fine Coir Pith + 20 % Perlite	55-65%	65-80%
Medium + Fibre (Dutch Plantin Classic)	85% Medium/fine Coir Pith + 15% Fibre	50-60%	60-75%
Sieved (Botanicoir Precision Plus)	90% Sieved Coir Pith (+2 mm) + 10% Fibre	45-55%	55-70%
Sieved + Chips (Botanicoir Precision Plus Ultra)	50% Sieved Coir Pith + 50% Chips (2-5 mm)	40-50%	50-65%

Different Coir substrates have widely different moisture retention – according to their physical make up.

- 10-15% increase in guideline moisture level after 1 seasons growth (2nd yr or L-C Pot ex. Cold Store)
- (Due to particle decomposition, root ingress & compaction).

Applying the Correct Level of Irrigation in Substrate

The correct level of irrigation in substrate is generally determined by 2 considerations.

1. Achieving the optimum root zone moisture level

- (more accurately - the optimum soil moisture tension)
- **Defined as:** at or just below Container Capacity (-10 to -40 mb tension for fruiting berries).

2. Achieving sufficient % run off.

- The optimal % run off will vary according to purity of background water and crop type.

Target Root Zone Moisture (tension) and % run off levels should be adjusted for each crop:

- according to the plant type, growth stage (& substrate age/composition).

Crop Growth Stage	Root Zone Moisture Level	Target Run off Level (%)
Rooting		
Growing		
Flowering		
Harvest		

Run off in Containerised Crops?

For plants grown in small volumes of substrate, run off (excess irrigation) is a necessary evil.

1. **To compensate for differences in dripper output (uniformity)**
 - (Ensures substrate is uniformly moist – in all pots, across the whole irrigation valve).
2. **To compensate for differences in water use (between individual pots)**
 - (Due to variation in plant size, in-field climate, topography, pot sizes/substrate, etc).
3. **To minimize EC build up in substrate (due to accumulation of excess salts)**
4. (e.g. Sodium, Chloride, Sulphate and/or unused fertilizer nutrients).
5. **To flush out potentially harmful elements that accumulate from irrigation water/substrate**
 - (e.g. Silicon + Boron in water/Coir)
6. **To replace nutrients used in large amounts,**
 - (Ensures an even balance in feed elements throughout entire substrate volume. Closer to drip recipe)



Excess drainage (run off)

However, run off should be kept to a minimum to avoid negative consequences:

1. **Over-watering** (Causing Root dieback)
2. **Waste of water and fertilizers**
3. **Increased mud/algae and grass/weed growth in plant rows**
4. **Increased humidity build up (in tunnels)**
5. **Increased substrate decomposition** (due to microbial break-down of peat/coir particles).
6. **Increased environmental pollution** (in soil and in nearby water courses).
 - (Increases risk of non-compliance with Local Nitrate & Phosphorus Control Directives).

Raspberry Pots - Run Off Collection Point

Collection point must be representative of whole crop.

1. Same level (**not** on raised platform)
 - Raised pots have higher water demand - more airflow.
 - Leads to over-watering whole crop.
2. Several pots/bags (5-10, not just 1 or 2).
 - Larger sample gives better average
3. Ideally ,1 collection point per valve
 - (even if not used every day).

Run off should be measured at least 2x/day.

- Just before 1st irrigation of the day
- (driest point in irrigation cycle).
- Midday/early afternoon
- (during peak water demand period).

This allows corrections to be made to irrigation programme in good time.



Ideally, collect drip solution from 2 (or more) drippers.
If possible, exclude light from drip + drain collection vessels

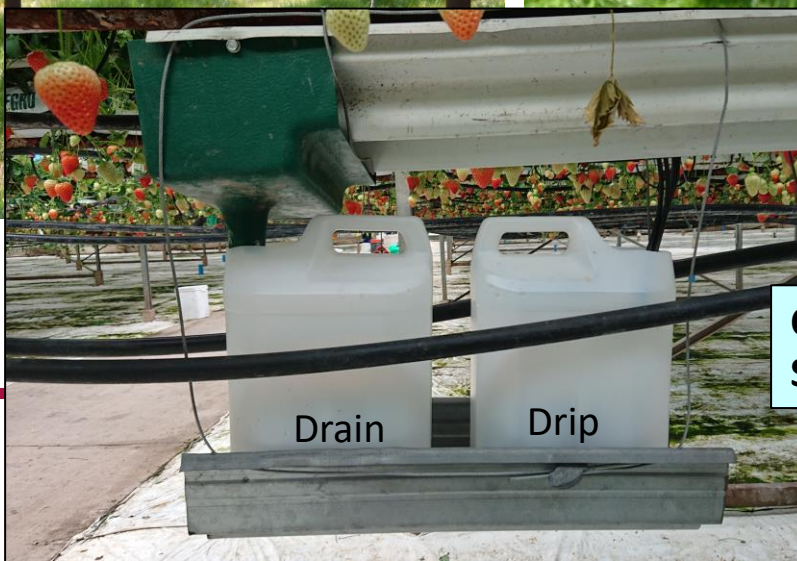
- (to minimise algae + consequent pH rise).

Strawberry Run Off Measuring Stations

- Simple (Manual) Run off Collecting Station



- Automatic Run Off Measuring Station



Combined Drip + Drain Collection Station (Glasshouse)

Calculation of % Run Off

A well set up run off monitoring station

- collects drip & drain samples from different numbers of drippers:

✦ **Formula for calculating % run off is:**

$$\% \text{ Run Off} = 100 \times \frac{\text{Volume of Run off}}{\text{Volume of H}_2\text{O in}} \times \frac{\text{No of Drippers (H}_2\text{O in)}}{\text{No of Drippers (Run Off)}}$$

Thus, if H₂O in (from 2 drippers) was 400 ml and volume of run off (from 9 drippers) was 330 ml:

$$\% \text{ run off} = 100 \times \frac{330}{400} \times \frac{2}{9} = \underline{\underline{18.33\%}}$$

Adjustment of Moisture and Run off at Different Growth Stages – Maravilla Gro-thru

Early pull outs of L-C in NW European Conditions (under Cold Glass/Tunnels)

Growth Stage	% Moisture*	% Run Off
Wet Up of Substrate (75% Coir Pith: 25% Fibre) →	45-50	15-25
Planting (Feb) →	50-55	20-30
Bud Break- Flowers (Mar-Apr) →	50-55	10-20
Flowers-1 st pick (Apr-Jun) →	55-65	20-30
Harvest on Floricane (Jun-July) →	50-60	15-25
Cut out of Floricane (mid July) →	45-65	5-35 (flushing?)
Primocane Flower- 1 st pick (July/August) →	55-65	20-30
Late (cold season) Harvest (Sept/Nov) →	50-60	5-15

* Suggested Moisture level in **Fresh Substrate** to keep approx -25 mb Tension (Plants re-potted).

- Original (nursery) substrate will be 10-15% higher.

Section 4

Crop Water Requirement and Irrigation Scheduling in Substrate



Worldwide Expertise for Food & Flowers

Understanding Hourly ET Demand (Crop Water Requirement in Substrate)

Hourly ET demand (vertical bars on chart)

- increases with increasing temperature.

Temperature & solar radiation (solid lines) are closely correlated.

- In summer, Solar Radiation is doubled
- ET demand rises proportionately.

In both seasons:

- 60-70% of ET demand is between 10.00 and 16.00.
- 7% of ET demand is in the dark period.
- (still some water demand at night).

ET demand is strongly correlated with VPD (dotted lines).

In Summer:

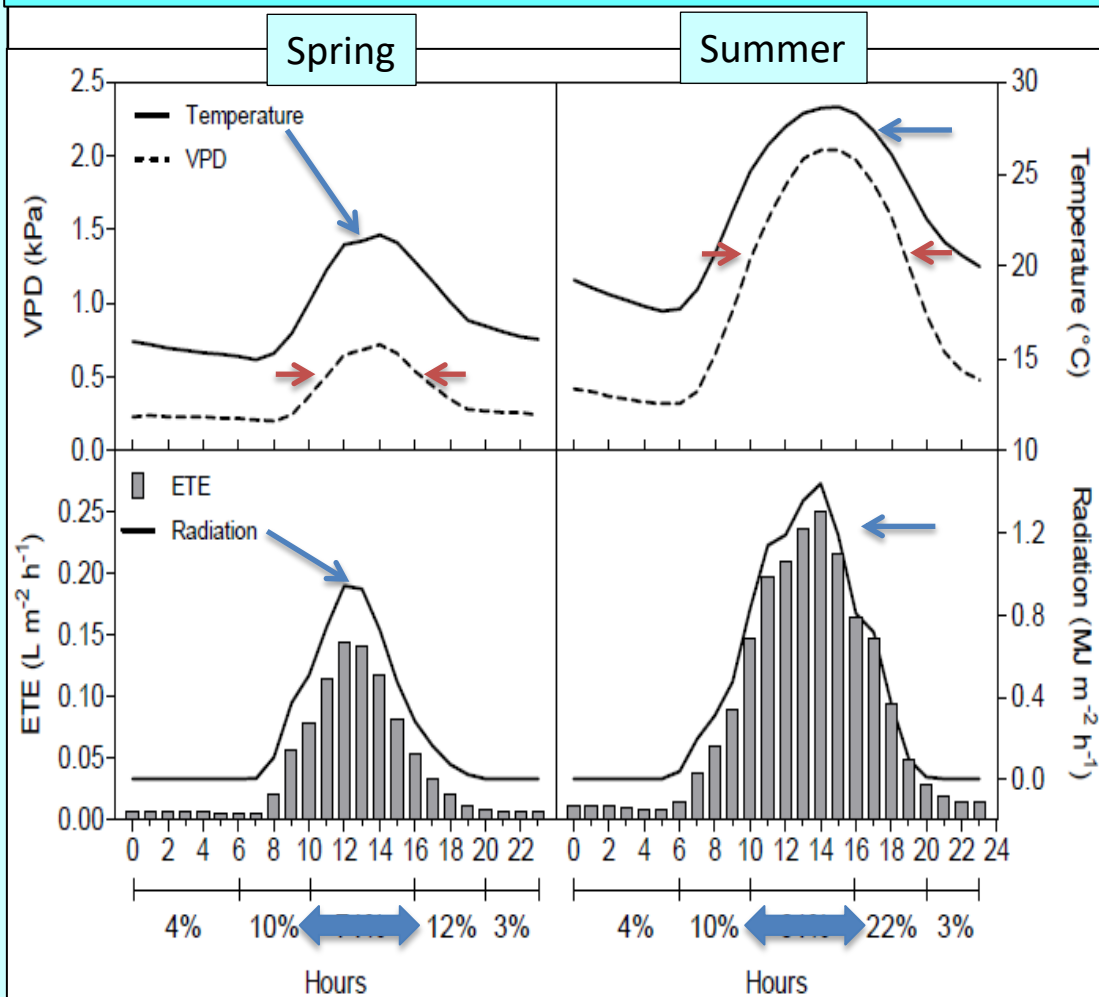
- VPD is generally above ideal levels
- (+1.2 kPa) between 10.00 and 16.00.

In Spring:

- VPD is below optimal levels (+0.5 kPa)
- from 16.00 to 10.00 (cool nights).

Hourly ET Demand of Crop grown in unheated tunnel

- Early spring (10 hrs) & mid-summer (15 hrs) daylight.



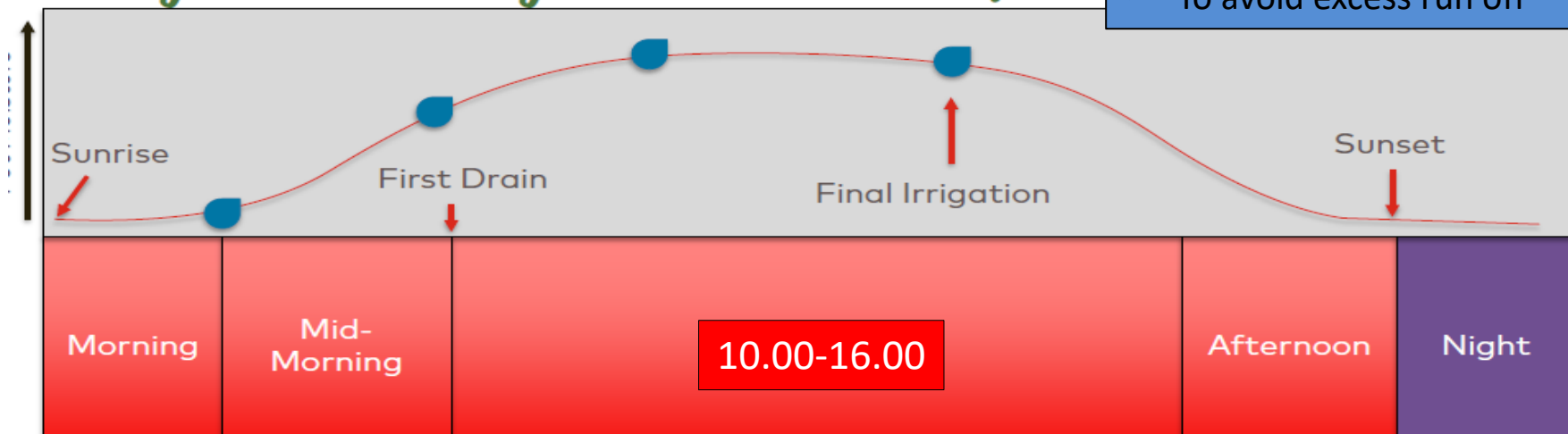
The 6 “Golden Rules” of Dripping Substrate Crops:

1. **Start dripping 1-2 hrs after sunrise and finish 1-3 hrs before sunset.**
2. **When there is little/no transpiration (e.g. at night), the substrate is allowed to dry back (a little)**
 - to introduce more air into the root zone.
3. **Almost every drip needs some run off (2-5-10-20-30%)**
 - (not the last drip giving 90%... pumping up the run off % for the day).
4. **If the first drip in the morning gave run off, then either:**
 - First drip was too early
 - Or, too much water was given the previous day
 - (i.e. too late, or too much volume).
 - **Aim to start Run Off during second drip of the morning.**
5. **Apply water “little & often” (to avoid over wetting the substrate & drying out between drip cycles).**
 - Aim to apply 1-2% of container volume/cycle in Strawbs & Blueberries
 - Aim to apply 2-4% of container volume in Rasp & Blackberries (due to faster rates of water uptake).
6. **On hot days, cluster drips around peak water use time (mid/late morning to late afternoon).**
 - Generally, around 65% of ET demand occurs between 10.00 and 16.00.

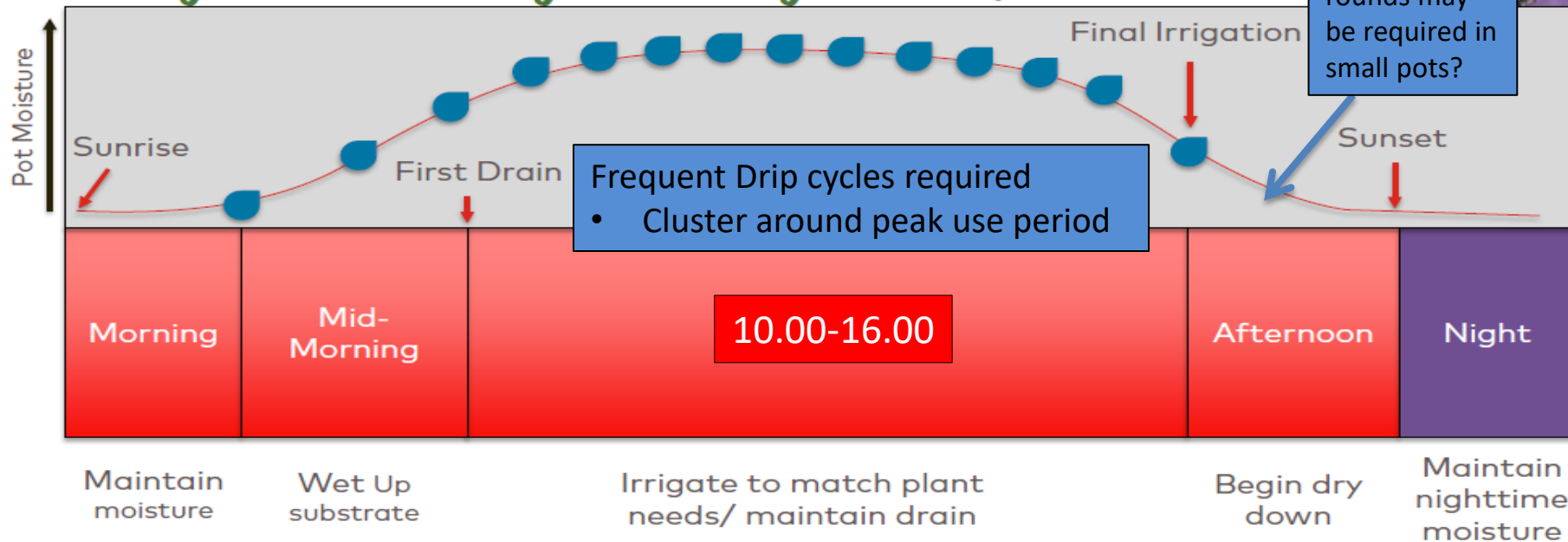


Irrigation Strategy - Small Canopy

Infrequent Drip Cycles required
 • To avoid excess run off



Irrigation Strategy - Large Canopy



Additional rounds may be required in small pots?

Frequent Drip cycles required
 • Cluster around peak use period

Maintain moisture

Wet Up substrate

Irrigate to match plant needs/ maintain drain

Begin dry down

Maintain nighttime moisture

● = Irrigation Event

Irrigation Scheduling Methods in Substrate?

Aim: to keep substrate evenly moist during the day by spreading out irrigation starts to match hourly ET demand.

1. Timed Starts. (Low tech option)

- Needs good knowledge of ET demand (water use) at different growth stages.
- Requires frequent alteration to timer schedule to cope with changes in weather.
- Least Accurate method. (v labour intensive in variable weather)

2. Weather Station. (Medium Tech Option).

- Can use accumulated Solar Radiation (Radsum) to track ET demand.
- Relatively close correlation for unheated crops grown under protection.
- Some growers use Vapour Pressure Deficit (VPD) – estimates ET potential from temp & humidity.
- Both methods take no account of wind or substrate moisture content (extra cycles may be required).

3. Root Zone Sensors. (High tech option).

- Use 1-5 fixed sensors, measuring either volumetric soil water content (VMC), or soil moisture tension.
- More expensive & VMC needs initial guesswork to set up full point (Container Capacity) of the substrate.
- Once set up, v accurate but may not be representative of whole field – due to small sample size.

4. Weighing of Crop (e.g. Priva Optimizer). Medium/High tech option.

- Selected area of crop is set up on weighing scale & irrigated according to change in weight (water uptake).
- (Priva system = v expensive). Best suited to hanging crops, where there is no ground level support.
- Again, may not be representative of whole crop.

Due to the inherent variability within many fields, it is important to cross-check with:

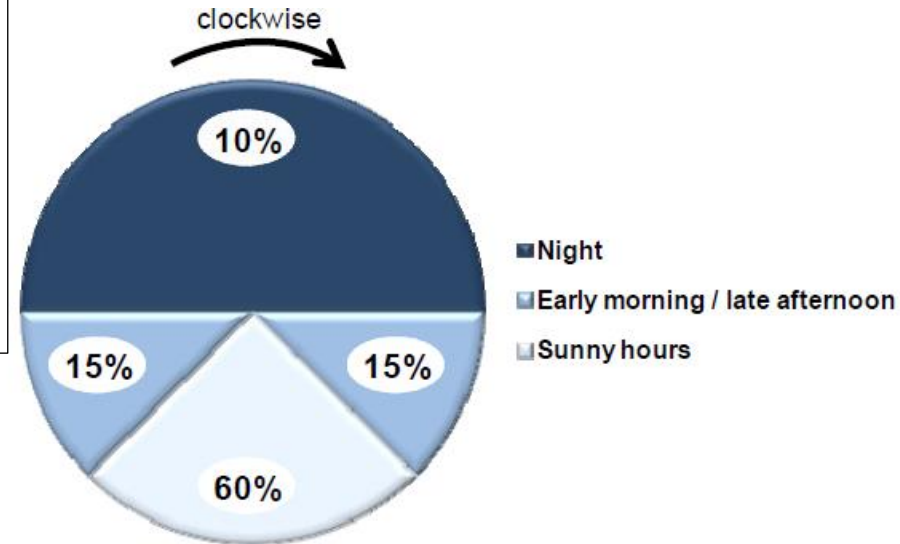
- **Fixed Run off measuring points** (1 per valve)
- **Daily roving assesment** with Sensor/Squeeze Tests across each whole valve.

Irrigation Scheduling Using Timer

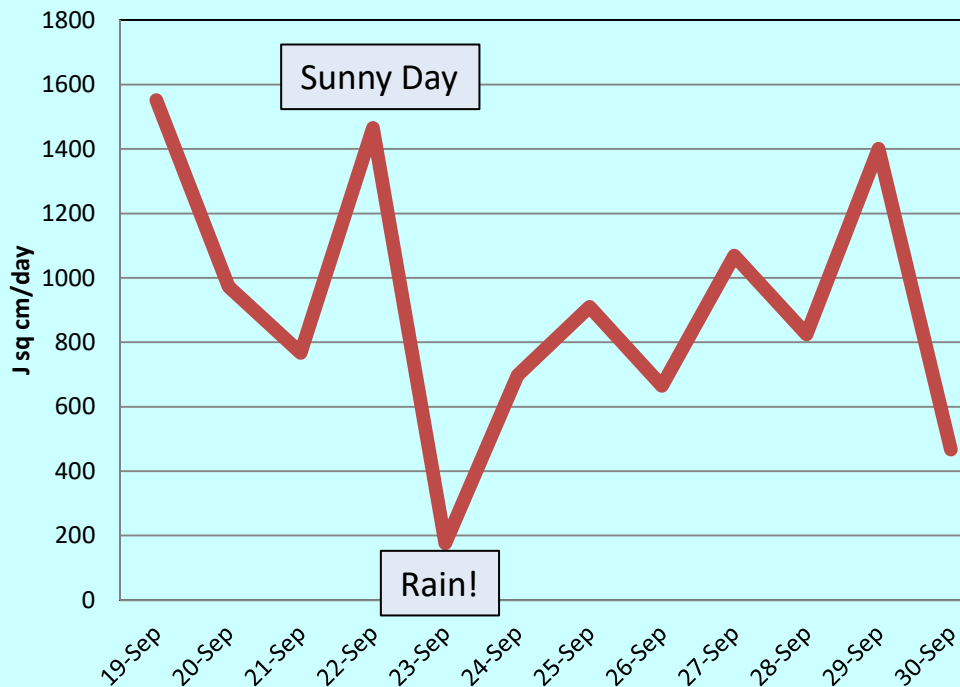
Aim: schedule water to meet hourly ET demand

Rough Guide:

- Apply 60-70% during peak hrs (10.00-16.00)
- 15% between sun rise & 10.00
- 15% from 16.00 to Sunset
- 10% at night (in summer). Not if cold/rainy!



**Ave Daily Radiation (W Sussex)
September 2012**



- In N Europe, can get 8x variation
- in Solar Radiation from one day to the next.
- Water demand will vary by a similar amount
- from 1 day to next.
- On any individual day, can also get:
- periods of high light intensity (strong sunlight).
- & periods of low light intensity (thick cloud or rainfall).
- **Frequent manual adjustments are necessary**
- **To avoid over-watering/under-watering**

Irrigation Scheduling Using Accumulated Light (Radsum)

Use a Solarimeter to count Accumulated Light Energy (Radsum)

- from Sunrise to Sunset
- Alter time of 1st & last drip manually.
- Set the Radsum interval between irrigation cycles
- (and volume applied per cycle) according to:
- Leaf Area of Crop, Volume of substrate & dripper set up
- (i.e. number and output of drippers per container).

Example:

- **L-Cane Raspberries with 2-3x 2 litre/hr drippers per 5-10 litre pot**
- **Before bud-break:** No leaf area (No Transpiration).
- Use Radsum interval of 200-350J - to wet up pots/replace evaporation.
- **On fully grown crop:** Leaf Area is high – pots dry fast.
- Use Radsum interval of 100-150 J to keep pace with high ET Demand.

Volume given:

- 75-150 ml/dripper/cycle (according to LAI, pot size + no of canes).
- Aim for 150 ml pot/cycle before bud break; 350-400 ml/pot/cycle in full leaf.

Can also integrate timed watering's (e.g. max rest time of 2-4 hrs)

- to account for additional ET due to wind
- or additional EC flushing cycles to bring down high EC.

Solarimeter Units of Light Measurement

- **Instantaneous Intensity** = Watts m⁻².
- **Accumulated Energy** = Joules cm⁻².
- **Joules = Watts x Seconds**
- 100 Watts m⁻² for 1 hour = 36 J cm⁻²
- (100W x 60 min x 60 sec ÷ 10 000).



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Peak Water Consumption by Pot Grown (Floricane) Raspberry Crop in N Europe (July)

Extreme situation:

- Plants fully loaded with fruit + full leaf canopy.
- Day Temp = 32C; Night Temp = 20C
- Day length = 16.5 hrs; Radsum = 3000J cm⁻²
- Wind speed = 9 m sec⁻¹ (Beaufort 5; 20 mph)
- 5625 pots/ha (1.5 pots/m & 2.67 m row spacing)
- Each pot has 3 Floricanes (4.5 Floricanes/m)
- 25% Drain required (between 10.00 & 16.00).

Peak Water Consumption calculation:

- Total Water Use measured at 45 m³/day
- Equates to 8 litres/pot/day
- (70% is needed between 10.00 & 16.00).
- 70% of 8 lits = 5.6 litres in 6 hours (933 ml hr⁻¹)
- At peak times, irrigation system needs to supply:
- 0.933 lits x 5625 = 5.25 m³ per ha per hour).

For a 4.7 litre pot with 2x 2 litre/hr drippers/pot:

- drip cycle of 300 ml/pot/cycle (150ml/dripper)
- Peak requirement = 3.1 drip cycles/hr
- (933 ml/pot/hr ÷ 300 ml/cycle).

Total irrigation requirement = 27 x 300 ml cycles/day
(Radsum Interval between cycles = approx 110J

- (3000J ÷ 27 cycles)



For a 7 litre pot with 3x 2 litre/hr drippers/pot:

- drip cycle of 350 ml/pot/cycle (117ml/dripper)
- At 933 ml/pot per hour:
- needs 2.65 drip cycles/hr during peak demand.

(Total irrigation requirement = 23 drip cycles/day)
(Radsum interval between cycles = approx 130J)

Strawberry - Estimated Water Consumption According to Radsum, Growth Stage + LAI

Water amount (ml per m2/day)				
	till flower	green fruit	start pick	full pick
Rad sum (J cm2)	LAI 1	LAI 2	LAI 3	LAI >3
100	125	187.5	250	312.5
200	187.5	312.5	375	437.5
300	250	437.5	625	750
400	312.5	562.5	875	1000
500	437.5	750	1125	1275
600	562.5	937.5	1375	1550
700	687.5	1125	1625	1825
800	812.5	1312.5	1875	2100
900	937.5	1500	2125	2375
1000	1062.5	1687.5	2375	2650
1100	1187.5	1875	2625	2925
1200	1312.5	2062.5	2875	3200
1300	1437.5	2250	3125	3475
1400	1562.5	2437.5	3375	3750
1500	1687.5	2625	3625	4025
1600	1812.5	2812.5	3875	4300
1700	1937.5	3000	4125	4575
1800	2062.5	3187.5	4375	4850
1900	2187.5	3375	4625	5125
2000	2312.5	3562.5	4875	5400
2100	2437.5	3750	5125	5675
2200	2562.5	3937.5	5375	5950
2300	2687.5	4125	5625	6225
2400	2812.5	4312.5	5875	6500
2500	2937.5	4500	6125	6775
2600	3062.5	4687.5	6375	7050
2700	3187.5	4875	6625	7325
2800	3312.5	5062.5	6875	7600
2900	3437.5	5250	7125	7875
3000	3562.5	5437.5	7375	8150

Estimated Water Use by Table Top Strawberries

- grown at 8-10 plants/linear m (in tunnels).

Figures includes 25% for run off

- but no allowance for wind in excess of 5 m sec⁻¹.

Water use varies from:

- <200ml/m (20-25 ml/l plant) in dull conditions
- (<200J cm⁻²) at start of crop.
- to >8.0 litres/m (800-100 ml/plant) in extreme conditions (3000J cm⁻²)
- with full leaf canopy and heavy crop load.

Glasshouse Crops generally use less water, as:

- A shading screen may be deployed
- Less influence of wind/air movement
- Fogging, roof sprinkling, or low level sprinkling may also be used.
- Results in a higher canopy RH, so lower VPD.**

Calculation of Maximum Water usage in Substrate Grown (Table top) Strawberries (UK)

Assumptions:

- Crop full grown (LAI = +3); 10 plants/linear m
- Fully loaded with fruit
- Outside temp = 30°C; RH 50% (VPD = 20 mb, extreme)
- Accumulated Radsum = 2500 J cm⁻²d⁻¹.
- Wind: 9 m/s (Beaufort 5; 20 mph)
- 25% Run off included

Measured plant water usage = 8 ltr/m/day (800 ml/plant)

- 20 lit bag with 4 x 1.5 l/hr drippers & 4 minute drip cycle
- = 400 ml/m per drip cycle (100 ml/dripper x 4 drippers)
- Needs 20 drip cycles/day.
- 70% of water required between 10.00-16.00 (560 ml/plant)
- Requires 14 drip cycles in 6 hours (5600 ml ÷ 400 ml/cycle)
- **Peak water demand = 2.33 drip cycles/hr (14 cycles ÷ 6 hrs)**
- **(i.e. 1 cycle every 26 mins)**
- **Set Radsum Interval at 125 Joules to deliver this.**
- (2500J x 70% ÷ 6 hrs = 292J hr⁻¹ ÷ 2.33 cycles)

Glasshouse Crops – water use may be 25% less (higher RH + less wind)

- Could keep Radsum @ 125J & reduce volume per cycle to 75 ml?



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Peak Water Consumption by Pot Grown Blueberry Crop - UK

Example (extreme situation)

- Plants fully loaded with fruit
- Day Temp = 30C; Night Temp = 20C (Radiation = 2500 J cm⁻²)
- Wind speed = 4-5 Beaufort (5.5-11 m/sec)
- 4 000 pots/ha (90cm pot spacing + 2.75 m row centres)
- 25% Drain required (between 10.00 & 16.00).

Peak Water Consumption calculation:

- Total Water Use measured at 28 m³/ha⁻¹ day⁻¹.
- Equates to 7 litres/pot/day
- (70% is needed between 10.00 and 16.00).
- 70% of 7 lits = 4.91 litres in 6 hours
- (817 ml/pot per hour – peak demand).
- Overall, irrigation system needs to supply:
- $0.817 \text{ l} \times 4000 = 3.27 \text{ m}^3 \text{ ha}^{-1} \text{ hr}^{-1}$ (peak demand).

With 3 x 2 litre hr⁻¹ drippers/25 litre pot (6 lit pot⁻¹hr⁻¹)

- 817 ml hr⁻¹ = 8.17 minutes of irrigation hr⁻¹
- 4 minute drip cycle = 133ml/dripper (300 ml pot⁻¹ cycle⁻¹)
- needs 2.04 drip cycles/hr
- (29 minute drip cycle - at peak demand)
- **(Total irrigation requirement =18 drip cycles/day)**
- Equates to Radsum interval of 140 J (2500J ÷ 18)



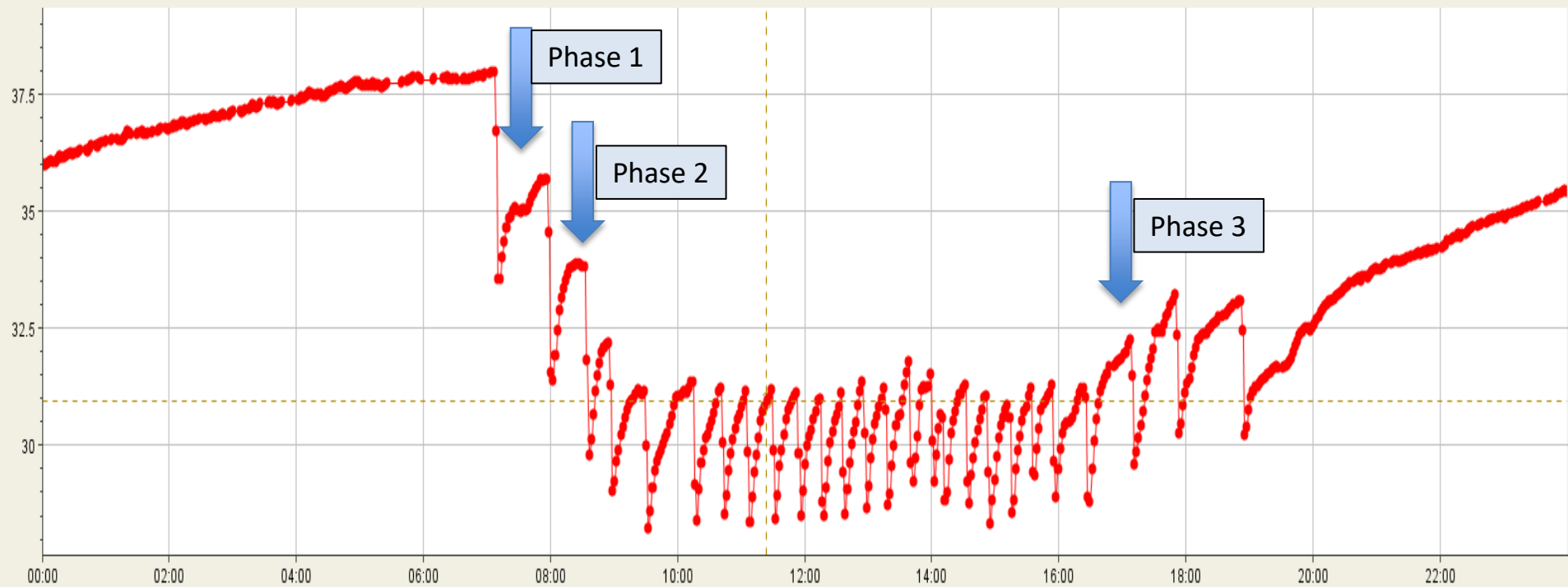
Large (vegetative) Draper Blueberry
in 25 litre round pot

Irrigation Scheduling Using Micro-Tensiometer

Some growers schedule by connecting an Electronic Micro-Tensiometer to their Irrigation controller. In the example below (Strawberries), a 3-phase Irrigation strategy is used. (Common Strategy in substrate):

- **Phase 1 – Early am (Managed wet up).** Manual start 2 hrs after sunrise, then hourly to 8.30 am.
- **Phase 2 – Peak Demand** (8.30 am to 4.30pm). Tension set at 31 mb to trigger irrigation starts.
- **Phase 3 – Late pm (Managed Dry-back).** From 5 pm to 2 hrs before sunset. Tension raised to 33 mb.

Analog signals 01-06-19 00:00:00 - 01-06-19 23:59:59



Volume given per cycle = 40 ml x 4 drippers/m = 160 ml per 1 m coir bag.

- 26 drip cycles given = 4.16 litres/m bag (x 8 plants = 520 ml/plant)

N.B. Crop LAI was 2.4; Radsum for the day = 2000J; Sunrise = 04.24, Sunset = 21.50.

Thanks for your attention

Questions?



Estimating Frequency of Dripping (from light Intensity) when Using Radsum

Radsum vs Light Intensity Irrigation Table

(Interval between irrigation cycles - minutes)



Light Intensity (Watts m ⁻²)	100	200	300	400	500	600	700	800	900	1000
Incident Energy* (J cm ⁻² hr ⁻¹)	36	72	108	144	180	216	252	288	324	360
Radsum (Jcm ⁻¹)										
75	125	63	42	31	25	21	18	16	14	13
100	167	83	56	42	33	28	24	21	19	17
125	208	104	69	52	42	35	30	26	23	21
150	250	125	83	58	50	42	38	31	28	25
175	292	146	97	73	58	49	42	36	32	29
200	333	167	111	83	67	56	48	42	37	33
225	375	188	125	94	75	58	54	47	42	38
250	417	208	139	104	83	69	60	52	46	42
275	458	229	151	115	93	76	66	57	51	46
300	500	250	167	125	100	83	71	63	56	50
325	542	271	181	135	108	90	77	68	60	54
350	583	292	194	146	117	97	83	73	65	58
400	667	333	222	167	133	111	95	83	74	67

* 1 watt = 1 Jsec⁻¹. So, 100 watts m⁻² for 1 hr = 100 x 60 x 60 J m⁻² Divide by 10 000 to get J cm⁻².

Moderate Light Intensity = <350 (<126 J cm⁻² hr⁻¹) = 1 drip cycle/hr at 125J Radsum Interval.

Bright Light Intensity = 350-700 Watts m⁻² (126-252 J cm⁻² hr⁻¹) = 1-2 cycles/hr at 125J Radsum Interval

V High Light Intensity = 700-1000+ Watts m⁻² (252-360+ J cm⁻² hr⁻¹) = 2-3 cycles/hr at 125J Radsum Interval.



Water Use in High Light Situation (Maravilla Gro-Thru in S Africa)

Typical Summer Day

- Temp = 30C; Radsum 3000 J Sq cm
- Peak Light Intensity = 900 W sq m (324 J sq cm/hr)

Growing Set Up

- 1 pot per m with 3 x 2 litre/hr drippers
- 4-5 Floricanes (+ 4-6+ Primocanes) per pot
- Row spacing = 2.5 m (4000 pots/ha)
- Grown under Light Diffusing Poly
- Average daily water use = 8 litres/pot (inc run off)
- Dripping @ 400 ml/pot/cycle (4 mins/hit)
- Overall need is 20 drip cycles per day.
- 20 drip cycles divided by 3000 Joules = 150 Joule Radsum Interval.
- Needs 2.16 drip cycles per hour (every 28 mins)

No Solarimeter available

- So, drip 2x/hr from 9.00-4pm
- 1x/hr from 6.00-9.00 & from 4.00 to 7.00 pm
- (also gives 20 drip cycles).

